# Beam Structured Channel Estimation for HF Skywave Massive MIMO-OFDM Communications

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***Abstract*— In this paper, we investigate high frequency (HF) skywave massive multiple-input multiple-output (MIMO) communications with orthogonal frequency division multiplexing (OFDM) modulation. Based on the triple-beam (TB) based channel model and the channel sparsity in the TB domain, we propose a beam structured channel estimation (BSCE) approach. Specifically, we show that the space-frequency-time (SFT) domain estimator design for each TB domain channel element can be transformed into that of a low-dimensional TB domain estimator and the resulting SFT domain estimator is beam structured. We also present a method to select the TBs used for BSCE. Then we generalize the proposed BSCE by introducing window functions and a turbo principle to achieve a superior trade-off between complexity and performance. Furthermore, we present a low-complexity design and implementation of BSCE by exploiting the characteristics of the TB matrix. Simulation results validate the proposed theory and methods.**

***Index Terms*— Massive MIMO-OFDM, HF skywave communications, beam structured channel estimation, low-complexity implementation.**

## I. Introduction

THERE is a growing need for future wireless networks to provide continuous global service coverage encompassing remote and rural areas [1], [2]. High frequency (HF) communications, operated in the frequency range of 3 to 30 MHz, allows for long-range over-the-horizon transmission through ionospheric reflection, enabling worldwide coverage [3], [4]. Nevertheless, traditional point-to-point HF communications face challenges, such as limited spectrum resources and complex ionospheric conditions, resulting in low data rates and restricted usability.

As a critical technology for the fifth-generation cellular wireless communications and its future evolution, massive multiple-input multiple-output (MIMO), which deploys a large number of antennas at the base station (BS), provides unprecedented spatial degrees of freedom and can serve a number of users simultaneously in the same time-frequency resource, thus significantly enhancing the system spectrum and energy efficiencies. Recently, massive MIMO in combination with orthogonal frequency division multiplexing (OFDM) has been introduced into HF skywave communications in [5], where the asymptotically achievable sum-rates of minimum mean-squared error (MMSE) based uplink (UL) receiver and downlink (DL) precoder have been derived and it has been shown that massive MIMO can provide significant performance gain for HF skywave communications. In [6], a low-complexity beam domain robust precoder has been proposed for HF skywave massive MIMO communications in high Doppler scenarios. In [7], low-complexity Slepian transform based detectors exploit beam domain sparsity of HF skywave massive MIMO channels.

Accurate channel state information (CSI) is indispensable to designing the UL receiver and DL precoder and harvesting the significant performance gain brought by massive MIMO. Therefore, channel estimation has been widely investigated in terrestrial cellular massive MIMO and mmWave massive MIMO systems. As the numbers of antennas and users increase, conventional channel estimation methods, such as least squares (LS) and MMSE [8], will lead to an unaffordable computational complexity. Due to the limited number of local scatters and propagation paths, the channel in terrestrial massive MIMO and mmWave massive MIMO communications typically exhibits sparsity [9], [10], which enables many efficient channel estimation methods. In [11] and [12], the channel has been estimated with the help of angle information obtained by the proposed spatial rotation enhanced discrete Fourier transform (DFT) method while the angle-delay information has been acquired in [13] to help estimate the channel in wideband systems. In [14] and [15], channel parameters, i.e.,

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angle of arrival (AoA), angle of departure (AoD) and path delay, have been estimated by unitary ESPRIT to reconstruct the channel. In [16], [17], [18], and [19], tensor-based signal processing schemes have been utilized to estimate the channel parameters by exploiting the low-rank property of massive MIMO channels. Moreover, the channel sparsity in massive MIMO communications allows channel estimation to be formulated as a sparse signal recovery problem and can be solved by compressive sensing schemes [20], [21]. In [22], [23], [24], [25], [26], [27], [28], greedy-based algorithms reconstruct the channel by progressively identifying the sparse support and estimating the corresponding channel coefficients. By jointly exploiting the channel sparsity and low-rank structures, a two-stage compressive sensing method, including a matrix completion stage and a sparse recovery stage, has been developed in [29]. Besides, many statistical inference methods have also been developed to perform efficient sparse channel estimation. In massive MIMO systems, the channel sparsity in angle domain has been exploited for channel estimation, such as the expectation-maximization (EM) Gaussian-mixture approximate message passing (AMP) algorithm [30] and the turbo-orthogonal AMP algorithm [31]. For wideband massive MIMO-OFDM systems, the angle-delay domain channel sparsity has also been utilized to perform efficient channel estimation, such as the structured turbo compressive sensing algorithm [32], the hierarchical hybrid message passing [33], and information geometry approach [34]. Note that the channel in HF skywave massive MIMO communications also exhibits significant sparsity [5], which implies that the above channel estimation algorithms that take advantage of the channel sparsity can be extended and applied also to HF skywave massive MIMO communications.

Recently, channel estimation for HF skywave massive MIMO-OFDM communications has been investigated in [35], where a triple-beam (TB) based channel model is established from the physical characteristic of HF skywave channels, and the channel sparsity in TB domain has been revealed. Based on the channel model and the TB domain channel sparsity, a low-overhead phase shift pilot design and a channel estimation algorithm based on constrained Bethe free energy minimization have been proposed. However, the computational complexity of the channel estimation algorithm in [35] is still large due to the large number of iterations.

In this paper, we further investigate channel estimation for HF skywave massive MIMO-OFDM communications. We fully exploit the channel sparsity in the TB domain to design a low-complexity channel estimation algorithm. The main contributions are summarized as follows.

- Based on the proposed TB based channel model and the phased shift pilots given in [35], the equivalent TB domain channels of all UTs can be non-overlapping with dedicated pilot scheduling, and thus estimating the TB domain channels for all UTs boils down to estimating the sum of the equivalent TB domain channels for all UTs, from which the TB domain channels of all UTs can be acquired.
- We propose the beam structured channel estimation (BSCE) for HF skywave massive MIMO-OFDM communications. Specifically, we show that the SFT domain estimator for each TB domain channel element can be designed by optimizing a low-dimensional TB domain estimator. The resulting SFT domain estimator is shown to be beam structured. We also present a method to choose the TBs used for BSCE, where the selected TBs are related to the TB that each TB domain channel element occupies.
- We further generalize the proposed BSCE in two aspects. First, we introduce window functions into BSCE to tackle the inter-TB interference caused by the non-orthogonality among TBs. Second, to improve the performance of BSCE especially when fewer TBs are used to construct BSCE, we introduce an iterative "turbo" extension of BSCE using the output of each iteration as a priori information for the next iteration to obtain a more accurate estimate. Thus a good trade-off between performance and complexity can be achieved.
- We present a low-complexity design and implementation for the proposed BSCE by exploiting the structure of the TB matrix and the asymptotic orthogonality among TBs. Moreover, we show that the computation of the TB domain estimator involves only real-valued matrices, thus reducing the computational complexity.

The rest of the paper is organized as follows. In Section II, we present the system model for HF skywave massive MIMO-OFDM communications. In Section III, BSCE is proposed with the TB based channel model and is generalized by introducing window functions and a turbo principle. In Section IV, low-complexity design and implementation are presented. Simulation results are provided in Section V and the paper is concluded in Section VI.

*Notations*: The uppercase (lowercase) boldface letters denote matrices (vectors). The superscripts $(\cdot)^*$, $(\cdot)^{\mathrm{T}}$, $(\cdot)^{\mathrm{H}}$, $(\cdot)^\dagger$ and $\mathrm{tr}\{\cdot\}$ denote the conjugate, transpose, conjugated-transpose, Moore-Penrose inverse and matrix trace operations, respectively. $\bar{\jmath}=\sqrt{-1}$. $\|\mathbf{x}\|$ denotes the $\ell_2$-norm of $\mathbf{x}$. $|\Psi|$ denotes the cardinality of set $\Psi$. $\Psi_1\times\Psi_2$ denotes the Cartesian product of sets $\Psi_1$ and $\Psi_2$. $\lceil x\rceil$ denotes the smallest integer that is not less than $x$ while $\lfloor x\rfloor$ denotes the largest integer that is not greater than $x$. $\circ$ and $\otimes$ denote the Hadamard and the Kronecker product operators, respectively. $[\mathbf{x}]_a$ and $[\mathbf{x}]_{\mathcal{S}}$ denote the $a$-th element and elements with index set $\mathcal{S}$ of vector $\mathbf{x}$, respectively. $[\mathbf{X}]_{a,b}$, $[\mathbf{X}]_{:,b}$, $[\mathbf{X}]_{a:b,c:d}$, and $[\mathbf{X}]_{:,\mathcal{S}}$ denote the $(a,b)$-th element, the $b$-th colomn, rows $a$ to $b$ and columns $c$ to $d$, and columns with index set $\mathcal{S}$ of $\mathbf{X}$, respectively. $[\mathcal{S}]_a$ denotes the $a$-th element of an ordered set $\mathcal{S}$. $\mathbb{E}\{\cdot\}$ and $\mathrm{Cov}\{\cdot\}$ denote the statistical expectation and covariance, respectively. $\mathrm{diag}\{\mathbf{x}\}$ denotes the diagonal matrix with $\mathbf{x}$ along its diagonal. $\mathbf{0}$ denote all-zero vector or matrix. $\mathbf{I}_N$ denotes the identity matrix of dimension $N$.

## II. System Model

In this section, after presenting the system configuration of HF skywave massive MIMO-OFDM communications, we review the TB based channel model previously introduced in [35] and the channel sparsity in the TB domain.

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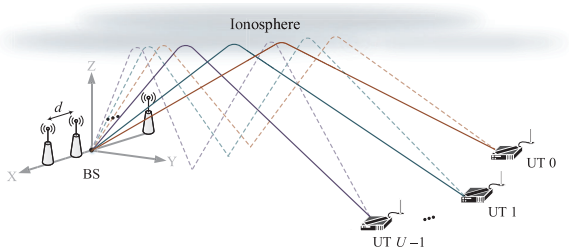



Fig. 1. Illustration of an HF skywave massive MIMO-OFDM system.

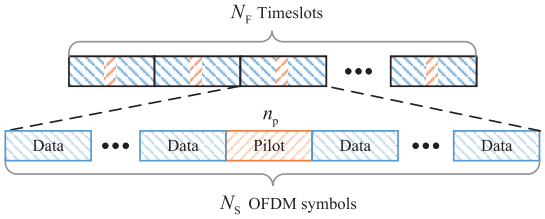



Fig. 2. The frame structure.

### A. System Configuration

We consider a long-range HF skywave massive MIMO-OFDM system as illustrated in Fig. 1, where the BS is equipped with a uniform linear array of $M$ antennas and provides long-range communications for $U$ single-antenna UTs.[1] Note that ULA is a reasonable configuration in the HF band and a typical example is the HF skywave over-the-horizon MIMO, where arrays deploy hundreds of antennas, leading to typically large antenna array aperture [36], [37], [38]. Since the ionospheric conditions in HF skywave communications may vary with time, the carrier frequency, $f_{\mathrm{c}}$, needs to be adjusted accordingly. Therefore, different from the conventional massive MIMO system where the inter-antenna spacing $d$ is typically set according to $f_{\mathrm{c}}$, we set inter-antenna spacing $d$ according to the highest used carrier frequency, $f_{\mathrm{o}}$, instead of carrier frequency $f_{\mathrm{c}}$, that is, $d = \lambda_{\mathrm{o}}/2$, where $\lambda_{\mathrm{o}} = c/f_{\mathrm{o}}$ is the wavelength, and $c$ is the speed of light.

We denote $N_{\mathrm{c}}$, $N_{\mathrm{g}}$, and $\Delta_f$ as the number of subcarriers, the length of cyclic prefix (CP), and the subcarrier spacing, respectively. Thus, the sampling interval length is $T_{\mathrm{s}} = 1/(N_{\mathrm{c}}\Delta_f)$ and the duration of each OFDM symbol with CP is $T_{\mathrm{sym}} = (N_{\mathrm{c}} + N_{\mathrm{g}})\,T_{\mathrm{s}}$. Within each OFDM symbol, $N_{\mathrm{v}}$ valid subcarriers are used for training and data transmission with the index set $\mathcal{K} = \{k_0, k_1, \cdots, k_{N_{\mathrm{v}}-1}\}$. The HF skywave massive MIMO-OFDM system is assumed to operate in time-division duplex (TDD) mode with frame structure shown in Fig. 2, where each frame consists of $N_{\mathrm{F}}$ timeslots and each timeslot contains $N_{\mathrm{S}}$ OFDM symbols, thus there are a total of $N = N_{\mathrm{F}}N_{\mathrm{S}}$ OFDM symbols in each frame. The $n_{\mathrm{p}}$-th OFDM symbol in each slot is a pilot used for UL training and the remaining OFDM symbols are used for UL/DL data transmission.

### B. Triple-Beam Based Channel Model

During the UL transmission, the channel frequency response at the $k$-th subcarrier on the $n$-th OFDM symbol between UT $u$ and the $m$-th antenna at the BS is given by [10], [35], and [39]

$$h^{\mathrm{SFT}}_{u,m,n,k} = \sum_{p=0}^{P_u-1} \gamma_{u,p} e^{\bar{\jmath}2\pi\nu_{u,p}nT_{\mathrm{sym}}} e^{-\bar{\jmath}2\pi f_{\mathrm{c}}m\Delta_\tau\Omega_{u,p}} e^{-\bar{\jmath}2\pi k\Delta_f\tau_{u,p}}, \tag{1}$$

where $P_u$ is the number of paths between the BS and UT $u$. In (1), $\gamma_{u,p} = \beta_{u,p}e^{\bar{\jmath}\varphi_{u,p}}$ is the complex-valued gain of the $p$-th path of UT $u$, where $\beta_{u,p}$ is the gain and $\varphi_{u,p}$ is the phase uniformly distributed over $[0, 2\pi)$. $\nu_{u,p}$ is the Doppler frequency caused by the UT mobility and the ionosphere. $\Omega_{u,p} = \sin\theta^{\mathrm{az}}_{u,p}\cos\theta^{\mathrm{el}}_{u,p}$ is the directional cosine, where $\theta^{\mathrm{az}}_{u,p}$ and $\theta^{\mathrm{el}}_{u,p}$ are the azimuth angle of arrival (AoA) and elevation AoA, respectively, and $\Delta_\tau = d/c$. $\tau_{u,p}$ is the $p$-th path delay between UT $u$ and the BS.

Note that the path parameters $(\Omega_{u,p}, \tau_{u,p}, \nu_{u,p})$ of each UT are limited within sets $\mathcal{B}_{\mathrm{an}} = \{\Omega\,|\Omega \in [-1, 1)\}$, $\mathcal{B}_{\mathrm{de}} = \{\tau\,|\tau \in [0, \tau_{\max})\}$, and $\mathcal{B}_{\mathrm{do}} = \{\nu\,|\nu \in [-\nu_{\max}, \nu_{\max})\}$, respectively, where $\tau_{\max} = N_{\mathrm{g}}/(N_{\mathrm{c}}\Delta_f)$ is the maximum delay spread and $\nu_{\max} = N_{\mathrm{d}}/(2NT_{\mathrm{sym}})$ is the maximum Doppler. We uniformly divide these sets into multiple disjoint subsets as [35]

$$\mathcal{B}_{\mathrm{an}} = \bigcup_{n_{\mathrm{an}}=0}^{N_{\mathrm{an}}-1} \Lambda^{\mathrm{an}}_{n_{\mathrm{an}}},\ \mathcal{B}_{\mathrm{de}} = \bigcup_{n_{\mathrm{de}}=0}^{N_{\mathrm{de}}-1} \Lambda^{\mathrm{de}}_{n_{\mathrm{de}}},\ \mathcal{B}_{\mathrm{do}} = \bigcup_{n_{\mathrm{do}}=0}^{N_{\mathrm{do}}-1} \Lambda^{\mathrm{do}}_{n_{\mathrm{do}}}, \tag{2}$$

where

$$\begin{aligned}
\Lambda^{\mathrm{an}}_{n_{\mathrm{an}}} &= [\Omega_{n_{\mathrm{an}}}, \Omega_{n_{\mathrm{an}}} + 2/N_{\mathrm{an}})\,,\\
\Lambda^{\mathrm{de}}_{n_{\mathrm{de}}} &= [\tau_{n_{\mathrm{de}}}, \tau_{n_{\mathrm{de}}} + N_\tau/(N_{\mathrm{de}}N_{\mathrm{v}}\Delta_f))\,,\\
\Lambda^{\mathrm{do}}_{n_{\mathrm{do}}} &= [\nu_{n_{\mathrm{do}}}, \nu_{n_{\mathrm{do}}} + N_{\mathrm{d}}/(N_{\mathrm{do}}NT_{\mathrm{sym}}))\,,
\end{aligned}$$

and $N_\tau = N_{\mathrm{v}}N_{\mathrm{g}}/N_{\mathrm{c}}$, $\Omega_{n_{\mathrm{an}}} = (2n_{\mathrm{an}} - N_{\mathrm{an}})/N_{\mathrm{an}}$, $\tau_{n_{\mathrm{de}}} = N_\tau n_{\mathrm{de}}/(N_{\mathrm{de}}N_{\mathrm{v}}\Delta_f)$, $\nu_{n_{\mathrm{do}}} = N_{\mathrm{d}}\,(n_{\mathrm{do}} - N_{\mathrm{do}}/2)/(N_{\mathrm{do}}NT_{\mathrm{sym}})$. The numbers of divided subsets for the directional cosine, the delay, and the Doppler frequency are $N_{\mathrm{an}} = \lceil F_{\mathrm{an}}Mf_{\mathrm{c}}/f_{\mathrm{o}}\rceil$, $N_{\mathrm{de}} = F_{\mathrm{de}}N_\tau$, and $N_{\mathrm{do}} = F_{\mathrm{do}}N_{\mathrm{d}}$, respectively, where $F_{\mathrm{an}}$, $F_{\mathrm{de}}$, and $F_{\mathrm{do}}$ are referred to as fine factors. Denote $\Gamma_u = \{(\Omega_{u,0},\tau_{u,0},\nu_{u,0}),(\Omega_{u,1},\tau_{u,1},\nu_{u,1}),\cdots,(\Omega_{u,P_u-1},\tau_{u,P_u-1},\nu_{u,P_u-1})\}$ and $\Lambda_{n_{\mathrm{an}},n_{\mathrm{de}},n_{\mathrm{do}}} = \Lambda^{\mathrm{an}}_{n_{\mathrm{an}}} \times \Lambda^{\mathrm{de}}_{n_{\mathrm{de}}} \times \Lambda^{\mathrm{do}}_{n_{\mathrm{do}}}$. Let $\mathbf{h}^{\mathrm{SFT}}_u \in \mathbb{C}^{MN_{\mathrm{v}}N}$ denote the space-frequency-time (SFT) domain channel between the BS and UT $u$, with $(nMN_{\mathrm{v}} + (k-k_0)M + m)$-th element, $h^{\mathrm{SFT}}_{u,m,n,k}$, then it can be expressed as [35]

$$\mathbf{h}^{\mathrm{SFT}}_u \approx \mathbf{P}\mathbf{h}^{\mathrm{TB}}_u, \tag{3}$$

where the $(n_{\mathrm{do}}N_{\mathrm{an}}N_{\mathrm{de}} + n_{\mathrm{de}}N_{\mathrm{an}} + n_{\mathrm{an}})$-th element of $\mathbf{h}^{\mathrm{TB}}_u$ is

$$\begin{aligned}
&\left[\mathbf{h}^{\mathrm{TB}}_u\right]_{n_{\mathrm{do}}N_{\mathrm{an}}N_{\mathrm{de}} + n_{\mathrm{de}}N_{\mathrm{an}} + n_{\mathrm{an}}}\\
&\quad = \sqrt{MN_{\mathrm{v}}N_{\mathrm{F}}}\left(\omega^{\mathrm{an}}_{n_{\mathrm{an}}}\omega^{\mathrm{de}}_{n_{\mathrm{de}}}\omega^{\mathrm{do}}_{n_{\mathrm{do}}}\right)^* \sum_{(\Omega_{u,p},\tau_{u,p},\nu_{u,p})\in\Gamma_u\cap\Lambda_{n_{\mathrm{an}},n_{\mathrm{de}},n_{\mathrm{do}}}} \gamma_{u,p},
\end{aligned} \tag{4}$$

and $\omega^{\mathrm{an}}_{n_{\mathrm{an}}} = e^{\bar{\jmath}2\pi\frac{M-1}{2}f_{\mathrm{c}}\Delta_\tau\Omega_{n_{\mathrm{an}}}}$, $\omega^{\mathrm{de}}_{n_{\mathrm{de}}} = e^{\bar{\jmath}2\pi\left(\frac{N_{\mathrm{v}}-1}{2}+k_0\right)\Delta_f\tau_{n_{\mathrm{de}}}}$, $\omega^{\mathrm{do}}_{n_{\mathrm{do}}} = e^{-\bar{\jmath}2\pi\left(\frac{(N_{\mathrm{F}}-1)N_{\mathrm{S}}}{2}+n_{\mathrm{p}}\right)T_{\mathrm{sym}}\nu_{n_{\mathrm{do}}}}$. In (3), $\mathbf{P}$ is given by

$$\mathbf{P} = \mathbf{D}\otimes\mathbf{U}\otimes\mathbf{V} \in \mathbb{C}^{MN_{\mathrm{v}}N\times N_{\mathrm{an}}N_{\mathrm{de}}N_{\mathrm{do}}}, \tag{5}$$

[1]The proposed channel estimation approach can also be applied to generic massive MIMO communications with different carrier frequencies and different system configurations although we focus on the HF skywave massive MIMO communications in this paper.

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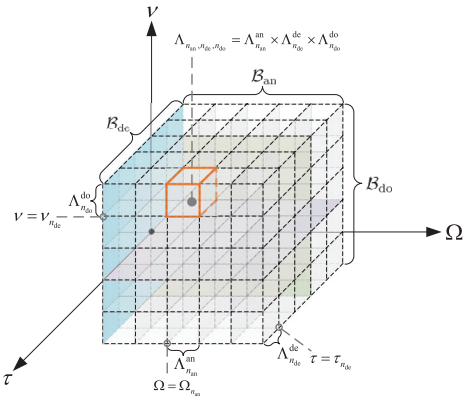



Fig. 3. Illustration of the TB domain.

where the $n_{\rm an}$-th column of $\mathbf{V} \in \mathbb{C}^{M\times N_{\rm an}}$, the $n_{\rm de}$-th column of $\mathbf{U} \in \mathbb{C}^{N_{\rm v}\times N_{\rm de}}$, and the $n_{\rm do}$-th column of $\mathbf{D} \in \mathbb{C}^{N\times N_{\rm do}}$ are respectively defined as

$$\mathbf{v}\left(\Omega_{n_{\rm an}}\right) \triangleq \frac{\omega^{\rm an}_{n_{\rm an}}}{\sqrt{M}}\Big[1, e^{-\bar{\jmath}2\pi f_{\rm c}\Delta\tau\Omega_{n_{\rm an}}}, \cdots \cdots, e^{-\bar{\jmath}2\pi f_{\rm c}(M-1)\Delta\tau\Omega_{n_{\rm an}}}\Big]^{\rm T} \in \mathbb{C}^{M}, \tag{6}$$

$$\mathbf{u}\left(\tau_{n_{\rm de}}\right) \triangleq \frac{\omega^{\rm de}_{n_{\rm de}}}{\sqrt{N_{\rm v}}}\Big[e^{-\bar{\jmath}2\pi k_0\Delta_f\tau_{n_{\rm de}}}, e^{-\bar{\jmath}2\pi k_1\Delta_f\tau_{n_{\rm de}}}, \cdots \cdots, e^{-\bar{\jmath}2\pi k_{N_{\rm v}-1}\Delta_f\tau_{n_{\rm de}}}\Big]^{\rm T} \in \mathbb{C}^{N_{\rm v}}, \tag{7}$$

$$\mathbf{d}(\nu_{n_{\rm do}}) \triangleq \frac{\omega^{\rm do}_{n_{\rm do}}}{\sqrt{N_{\rm F}}}\Big[1, e^{\bar{\jmath}2\pi\nu_{n_{\rm do}}T_{\rm sym}}, \cdots, e^{\bar{\jmath}2\pi\nu_{n_{\rm do}}(N-1)T_{\rm sym}}\Big]^{\rm T} \in \mathbb{C}^{N}, \tag{8}$$

which are steering vectors in the space domain, the frequency domain, and the time domain, pointing towards the sampled directional cosine $\Omega_{n_{\rm an}}$, delay $\tau_{n_{\rm de}}$, and Doppler frequency $\nu_{n_{\rm do}}$, respectively. Then each column of $\mathbf{P}$ is referred to as a TB, the matrix $\mathbf{P}$ is referred to as a TB matrix, $\mathbf{h}^{\rm TB}_u$ is the TB domain channel and (3) is referred to as the TB based channel model [35].[2] Note that the introduction of the phase shift coefficients, $\omega^{\rm an}_{n_{\rm an}}$, $\omega^{\rm an}_{n_{\rm an}}$, $\omega^{\rm an}_{n_{\rm an}}$, makes the steering vectors in (6)-(8) exhibit conjugate centrosymmetry, which can be used to reduce the computational complexity of BSCE proposed subsequently, which will be discussed in detail in Sec. IV-A.

Moreover, define the TB domain covariance matrix of UT $u$, also referred to as the TB domain statistical CSI, as

$$\mathbf{R}^{\rm TB}_u = \mathbb{E}\left\{\mathbf{h}^{\rm TB}_u\left(\mathbf{h}^{\rm TB}_u\right)^{\rm H}\right\}. \tag{9}$$

We assume that the TB domain channel coefficients, i.e., elements of $\mathbf{h}^{\rm TB}_u$, follow independent complex Gaussian distributions with zero mean and different variances. Thus $\mathbf{R}^{\rm TB}_u$ is a diagonal matrix with $(n_{\rm do}N_{\rm an}N_{\rm de}+n_{\rm de}N_{\rm an}+n_{\rm an})$-th diagonal element $MN_{\rm v}N_{\rm F}\sum_{(\Omega_{u,p},\tau_{u,p},\nu_{u,p})\in\Gamma_u\cap\Lambda_{n_{\rm an},n_{\rm de},n_{\rm do}}}\beta^2_{u,p}$. In HF skywave massive MIMO-OFDM communications, due to the limited number of propagation paths, Doppler spread, and small angle spread, the channel typically exhibits sparsity in the TB domain, which means that most elements of the TB domain channels are approximately zero. Moreover, note that the HF skywave channel parameters can be nearly stationary for a long time relative to the communication timescale [37], [41]. Due to the channel sparsity in the TB domain and the slow-varying characteristic of the statistical CSI [13], [42], [43], we assume that $\mathbf{R}^{\rm TB}_u$ is available at the BS in the rest of the paper.

[2]Note that the angle-delay-Doppler domain channel in the literatures [10], [18], and [40] is similar to the TB domain channel.

## III. Beam Structured Channel Estimation

In this section, we investigate the channel estimation with the TB based channel model. We show that the SFT domain estimator is beam structured and can be designed by optimizing a low-dimensional TB domain estimator. How to select TBs used for such a BSCE is also discussed. Moreover, we generalize the proposed BSCE by introducing window functions and a turbo principle to make a good trade-off between complexity and performance.

### A. Problem Formulation

We consider the pilot based channel estimation. According to the frame structure in Fig. 2, the SFT domain channel at pilot segments is given by [3]

$$\mathbf{h}^{\rm SFT,p}_u = \left([\mathbf{D}]_{\mathcal{P},:}\otimes\mathbf{U}\otimes\mathbf{V}\right)\mathbf{h}^{\rm TB}_u \in \mathbb{C}^{MN_{\rm v}N_{\rm F}}, \tag{10}$$

where $\mathcal{P} = \{n_{\rm p}, n_{\rm p}+N_{\rm S}, \cdots, n_{\rm p}+(N_{\rm F}-1)N_{\rm S}\}$ is the index set of pilots in the frame. Then the corresponding frequency domain received signal at the BS is

$$\mathbf{y} = \sum_{u=0}^{U-1}\mathbf{X}_u\mathbf{h}^{\rm SFT,p}_u + \mathbf{z} = \mathbf{X}\mathbf{h}^{\rm SFT,p} + \mathbf{z}, \tag{11}$$

where $\mathbf{h}^{\rm SFT,p} = \left[\left(\mathbf{h}^{\rm SFT,p}_0\right)^{\rm T}, \left(\mathbf{h}^{\rm SFT,p}_1\right)^{\rm T}, \cdots, \left(\mathbf{h}^{\rm SFT,p}_{U-1}\right)^{\rm T}\right]^{\rm T}$, $\mathbf{X} = [\mathbf{X}_0, \mathbf{X}_1, \cdots, \mathbf{X}_{U-1}]$, $\mathbf{X}_u = \mathbf{I}_{N_{\rm F}}\otimes(\mathrm{diag}\{\mathbf{x}^{\rm p}_u\}\otimes\mathbf{I}_M)$, $\mathbf{x}^{\rm p}_u \in \mathbb{C}^{N_{\rm v}}$ denotes the transmitted pilot of UT $u$ at valid subcarriers, and $\mathbf{z}\in\mathbb{C}^{MN_{\rm v}N_{\rm F}}$ is the additive white Gaussian noise with distribution $\mathcal{CN}\left(\mathbf{0}, \sigma^2_{\rm z}\mathbf{I}_{MN_{\rm v}N_{\rm F}}\right)$.

We choose the phase shift pilots in [35], which is given by

$$\mathbf{x}^{\rm P}_u = \sigma_{\rm p}\mathbf{x}_{\rm c}\circ\left[e^{-\bar{\jmath}2\pi k_0\frac{N_\tau\phi_u}{N_{\rm de}N_{\rm v}}}, e^{-\bar{\jmath}2\pi k_1\frac{N_\tau\phi_u}{N_{\rm de}N_{\rm v}}}, \cdots, e^{-\bar{\jmath}2\pi k_{N_{\rm v}-1}\frac{N_\tau\phi_u}{N_{\rm de}N_{\rm v}}}\right]^{\rm T}, \tag{12}$$

where $\sigma_{\rm p}$ is the square root of the pilot transmit power, $\phi_u \in \{0, N_{\rm de}, \cdots, (\lfloor N_{\rm v}/N_\tau\rfloor - 1)N_{\rm de}\}$ is referred to as the phase shift factor and $\mathbf{x}_{\rm c}$ is a sequence with unit modulus elements shared by all UTs. A suitable option for $\mathbf{x}_{\rm c}$ is the Zadoff-Chu (ZC) sequence [44], which has been widely employed in terrestrial cellular wireless networks. Thus according to [35], to minimize the normalized mean-squared error (NMSE) of the channel estimate, UTs with overlapping TB domain channels should be allocated pilots with different phase shift factors while UTs with non-overlapping TB domain channels can reuse the same pilot. The degree of TB domain channel overlap between UTs $u$ and $u'$ can be quantified as

[3]Using the channel prediction method in [35], the SFT domain channel at data segments can also be obtained with the TB based channel model and the TB domain channel estimate acquired by the pilot based channel estimation.

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$\rho_{u,u'} = \mathrm{tr}\{\mathbf{R}_u^{\mathrm{TB}}\mathbf{R}_{u'}^{\mathrm{TB}}\}/(\|\mathbf{R}_u^{\mathrm{TB}}\|_{\mathrm{F}}\|\mathbf{R}_{u'}^{\mathrm{TB}}\|_{\mathrm{F}})$ as in [35], which can be thresholded to distinguish whether the TB domain channels of UTs are overlapping. Then we have

$$\begin{aligned}\mathbf{X}_u\mathbf{h}_u^{\mathrm{SFT}} &= \sigma_{\mathrm{p}}\mathbf{X}_{\mathrm{c}}\left([\mathbf{D}]_{\mathcal{P},:}\otimes\bar{\mathbf{U}}\otimes\mathbf{V}\right)\mathbf{S}_u\mathbf{h}_u^{\mathrm{TB}}\\ &= \sigma_{\mathrm{p}}\mathbf{X}_{\mathrm{c}}\hat{\mathbf{P}}\mathbf{S}_u\mathbf{h}_u^{\mathrm{TB}},\end{aligned} \tag{13}$$

where $\mathbf{X}_{\mathrm{c}} \triangleq \mathbf{I}_{N_{\mathrm{F}}}\otimes(\mathrm{diag}\{\mathbf{x}_{\mathrm{c}}\}\otimes\mathbf{I}_M)$, $\bar{\mathbf{U}} \in \mathbb{C}^{N_{\mathrm{v}}\times\bar{N}_{\mathrm{de}}}$ with $[\bar{\mathbf{U}}]_{:,n_{\mathrm{de}}} = \mathbf{u}(\tau_{n_{\mathrm{de}}})$, $\bar{N}_{\mathrm{de}} = \lfloor N_{\mathrm{v}}/N_\tau\rfloor N_{\mathrm{de}}$, $\mathbf{S}_u = \left(\mathbf{I}_{N_{\mathrm{do}}}\otimes\left[\mathbf{0}_{N_{\mathrm{de}}\times\phi_u},\mathbf{I}_{N_{\mathrm{de}}},\mathbf{0}_{N_{\mathrm{de}}\times((\lfloor N_{\mathrm{v}}/N_\tau\rfloor-1)N_{\mathrm{de}}-\phi_u)}\right]^{\mathrm{T}}\otimes\mathbf{I}_{N_{\mathrm{an}}}\right)$, and

$$\hat{\mathbf{P}} \triangleq [\mathbf{D}]_{\mathcal{P},:}\otimes\bar{\mathbf{U}}\otimes\mathbf{V} \in \mathbb{C}^{MN_{\mathrm{v}}N_{\mathrm{F}}\times N_{\mathrm{an}}\bar{N}_{\mathrm{de}}N_{\mathrm{do}}}. \tag{14}$$

We refer to $\mathbf{S}_u\mathbf{h}_u^{\mathrm{TB}}$ as the equivalent TB domain channel of UT $u$. Since the channel is sparse in the TB domain, with proper pilot allocation as in [35], the equivalent TB domain channels of all UTs can be non-overlapping, as shown in Fig. 4. This is since that when UTs with overlapping TB domain channels are assigned pilots with different phase shift factors, their equivalent TB domain channel will exhibit corresponding shift along the delay-dimensional, thereby being non-overlapping in the TB domain [43], [45]. In this case, (11) can be rewritten as

$$\mathbf{y} = \sigma_{\mathrm{p}}\mathbf{X}_{\mathrm{c}}\hat{\mathbf{P}}\sum_{u=0}^{U-1}\mathbf{S}_u\mathbf{h}_u^{\mathrm{TB}} + \mathbf{z} \approx \sigma_{\mathrm{p}}\mathbf{X}_{\mathrm{c}}\bar{\mathbf{P}}\bar{\mathbf{h}}^{\mathrm{TB}} + \mathbf{z}, \tag{15}$$

where $\bar{\mathbf{P}} = [\hat{\mathbf{P}}]_{:,\bar{\mathcal{N}}^{\mathrm{TB}}}$, $\bar{\mathbf{h}}^{\mathrm{TB}} = [\sum_{u=0}^{U-1}\mathbf{S}_u\mathbf{h}_u^{\mathrm{TB}}]_{\bar{\mathcal{N}}^{\mathrm{TB}}}$, $\bar{\mathcal{N}}^{\mathrm{TB}} = \bigcup_{u=0}^{U-1}\bar{\mathcal{N}}_u^{\mathrm{TB}}$ with $\bar{N}_{\mathrm{TB}} = |\bar{\mathcal{N}}^{\mathrm{TB}}|$, and $\bar{\mathcal{N}}_u^{\mathrm{TB}} = \{n_{\mathrm{TB}} | [\mathbf{S}_u\mathbf{R}_u^{\mathrm{TB}}\mathbf{S}_u^{\mathrm{H}}]_{n_{\mathrm{TB}},n_{\mathrm{TB}}} > \eta\}$ is the non-zero element index set of the equivalent TB domain channels of UT $u$, where $\eta$ is a small threshold used to filter out elements that are approximately zero. Here, $\bar{\mathcal{N}}^{\mathrm{TB}}$ is an ordered set, where the order is the index increasing order. Note that $\mathbf{X}_{\mathrm{c}}$ is a diagonal pilot matrix with unit modulus diagonal elements. To remove the effect of the pilot matrix, we define

$$\bar{\mathbf{y}} \triangleq \mathbf{X}_{\mathrm{c}}^{\mathrm{H}}\mathbf{y}. \tag{16}$$

Note that this is without loss of optimality of the channel estimation since all pilot elements are unit magnitude and the distribution of the noise is not changed. In this case, the estimation of TB domain channels for all UTs is equivalent to estimating the sum of the equivalent TB domain channels for all UTs, i.e., $\bar{\mathbf{h}}^{\mathrm{TB}}$, as illustrated in Fig. 4. Speciafically, when the MMSE estimation is applied, the estimation result is

$$\bar{\mathbf{h}}_{\mathrm{MMSE}}^{\mathrm{TB}} = \sigma_{\mathrm{p}}\bar{\mathbf{R}}^{\mathrm{TB}}\left(\sigma_{\mathrm{p}}^2\bar{\mathbf{P}}^{\mathrm{H}}\bar{\mathbf{P}}\bar{\mathbf{R}}^{\mathrm{TB}} + \sigma_{\mathrm{z}}^2\mathbf{I}\right)^{-1}\bar{\mathbf{P}}^{\mathrm{H}}\bar{\mathbf{y}}, \tag{17}$$

where $\bar{\mathbf{R}}^{\mathrm{TB}} = [\sum_{u=0}^{U-1}\mathbf{S}_u\mathbf{R}_u^{\mathrm{TB}}\mathbf{S}_u^{\mathrm{H}}]_{\bar{\mathcal{N}}^{\mathrm{TB}},\bar{\mathcal{N}}^{\mathrm{TB}}}$. Then the estimation of the TB domain channels for UT $u$ is $\hat{\mathbf{h}}_u^{\mathrm{TB}} = [\mathbf{B}\bar{\mathbf{h}}_{\mathrm{MMSE}}^{\mathrm{TB}}]_{\bar{\mathcal{N}}_u^{\mathrm{TB}}}$, where $\mathbf{B} \in \mathbb{C}^{N_{\mathrm{an}}\bar{N}_{\mathrm{de}}N_{\mathrm{do}}\times\bar{N}_{\mathrm{TB}}}$ is defined as

$$[\mathbf{B}]_{i,j} = \begin{cases}1, & i = [\bar{\mathcal{N}}^{\mathrm{TB}}]_j,\\ 0, & \text{otherwise.}\end{cases} \tag{18}$$

Then the estimate of $\mathbf{h}_u^{\mathrm{SFT,p}}$ is $\hat{\mathbf{h}}_u^{\mathrm{SFT,p}} = [\mathbf{P}]_{:,\mathcal{N}_u^{\mathrm{TB}}}\hat{\mathbf{h}}_u^{\mathrm{TB}}$, where $\mathcal{N}_u^{\mathrm{TB}} = \{n_{\mathrm{TB}} | [\mathbf{R}_u^{\mathrm{TB}}]_{n_{\mathrm{TB}},n_{\mathrm{TB}}} > \eta\}$. Note that the complexity of the MMSE estimation in (17) is $\mathcal{O}\left(\bar{N}_{\mathrm{TB}}^2MN_{\mathrm{v}}N_{\mathrm{F}} + \bar{N}_{\mathrm{TB}}^3\right)$. Clearly for this MMSE estimation, when the number of UTs becomes large, the large $\bar{N}_{\mathrm{TB}}$ will make the computation of $\bar{\mathbf{h}}_{\mathrm{MMSE}}^{\mathrm{TB}}$ unaffordable in practical systems due to large-dimensional matrix inversion and multiplication.

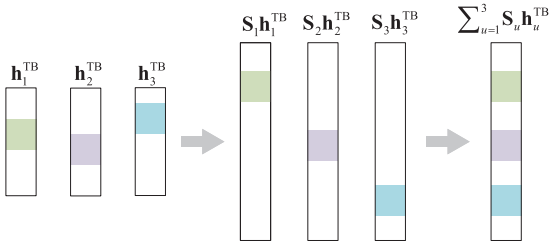



Fig. 4. Illustration of equivalent TB domain channels.

### B. Beam Structured Channel Estimation

We consider the estimator design for the $\bar{n}$-th element of $\bar{\mathbf{h}}^{\mathrm{TB}}$, which is defined by a weight vector $\mathbf{w}_{\bar{n}} \in \mathbb{C}^{MN_{\mathrm{v}}N_{\mathrm{F}}}$ to the received signal $\bar{\mathbf{y}}$ in (16), and the corresponding mean-squared error (MSE) is given by

$$\begin{aligned}\mathrm{MSE}_{\bar{n}} &= \mathbb{E}\left\{\left\|[\bar{\mathbf{h}}^{\mathrm{TB}}]_{\bar{n}} - \mathbf{w}_{\bar{n}}^{\mathrm{H}}\bar{\mathbf{y}}\right\|^2\right\}\\ &= [\bar{\mathbf{R}}^{\mathrm{TB}}]_{\bar{n},\bar{n}} - \sigma_{\mathrm{p}}[\bar{\mathbf{R}}^{\mathrm{TB}}]_{\bar{n},\bar{n}}\left(\mathbf{w}_{\bar{n}}^{\mathrm{H}}\bar{\mathbf{P}}\mathbf{e}_{\bar{n}} + \mathbf{e}_{\bar{n}}^{\mathrm{H}}\bar{\mathbf{P}}^{\mathrm{H}}\mathbf{w}_{\bar{n}}\right)\\ &\quad + \mathbf{w}_{\bar{n}}^{\mathrm{H}}\left(\sigma_{\mathrm{p}}^2\bar{\mathbf{P}}\bar{\mathbf{R}}^{\mathrm{TB}}\bar{\mathbf{P}}^{\mathrm{H}} + \sigma_{\mathrm{z}}^2\mathbf{I}\right)\mathbf{w}_{\bar{n}},\end{aligned} \tag{19}$$

where $\mathbf{e}_{\bar{n}} = [\mathbf{I}_{\bar{N}_{\mathrm{TB}}}]_{:,\bar{n}}$. Then the design of estimator $\mathbf{w}_{\bar{n}}$ is formulated as the following problem

$$\mathbf{w}_{\bar{n}}^{\mathrm{op}} = \arg\min_{\mathbf{w}_{\bar{n}}}\ \mathrm{MSE}_{\bar{n}}, \tag{20}$$

and we have the following lemma, proved in Appendix.

*Lemma 1:* The optimal solution of problem (20) can be obtained as

$$\mathbf{w}_{\bar{n}}^{\mathrm{op}} = \bar{\mathbf{P}}\left(\bar{\mathbf{P}}^{\mathrm{H}}\bar{\mathbf{P}}\right)^{\dagger}\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}, \tag{21}$$

where

$$\begin{aligned}\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{op}} = \underset{\tilde{\mathbf{w}}_{\bar{n}}}{\mathrm{argmin}}\Big\{&[\bar{\mathbf{R}}^{\mathrm{TB}}]_{\bar{n},\bar{n}} - \sigma_{\mathrm{p}}[\bar{\mathbf{R}}^{\mathrm{TB}}]_{\bar{n},\bar{n}}\left(\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{H}}\mathbf{e}_{\bar{n}} + \mathbf{e}_{\bar{n}}^{\mathrm{H}}\tilde{\mathbf{w}}_{\bar{n}}\right)\\ &+ \tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{H}}\left(\sigma_{\mathrm{p}}^2\bar{\mathbf{R}}^{\mathrm{TB}} + \sigma_{\mathrm{z}}^2\left(\bar{\mathbf{P}}^{\mathrm{H}}\bar{\mathbf{P}}\right)^{\dagger}\right)\tilde{\mathbf{w}}_{\bar{n}}\Big\}\\ \text{s.t.}\ &\left(\bar{\mathbf{P}}^{\mathrm{H}}\left(\bar{\mathbf{P}}^{\mathrm{H}}\right)^{\dagger} - \mathbf{I}\right)\tilde{\mathbf{w}}_{\bar{n}} = \mathbf{0}.\end{aligned} \tag{22}$$

As we will see below, with the expression in Lemma 1, we are able to simplify the optimal estimator with a simple and closed form in an HF skywave massive MIMO-OFDM system. In the optimization problem in (22), we can turn to optimize $\tilde{\mathbf{w}}_{\bar{n}}$ instead of $\mathbf{w}_{\bar{n}}$ to minimize the MSE. When we obtain $\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}$ by solving optimization problem (22), we can directly construct estimator $\mathbf{w}_{\bar{n}}^{\mathrm{op}}$ according to (21). Moreover, we assume that the number of UTs is limited and the directional cosines of the paths of UTs are distinct [5]. Thus when $M, N_{\mathrm{v}}, N_{\mathrm{F}} \to \infty$, the columns of $\bar{\mathbf{P}}$ are asymptotically orthogonal, i.e.,

$$\lim_{M,N_{\mathrm{v}},N_{\mathrm{F}}\to\infty}\bar{\mathbf{P}}^{\mathrm{H}}\bar{\mathbf{P}} = \mathbf{I}. \tag{23}$$

In this case, it can be checked that $\lim_{M,N_{\mathrm{v}},N_{\mathrm{F}}\to\infty}\bar{\mathbf{P}}^{\mathrm{H}}\left(\bar{\mathbf{P}}^{\mathrm{H}}\right)^{\dagger} - \mathbf{I} = \mathbf{0}$ and $\lim_{M,N_{\mathrm{v}},N_{\mathrm{F}}\to\infty}\left(\bar{\mathbf{P}}^{\mathrm{H}}\bar{\mathbf{P}}\right)^{\dagger} = \mathbf{I}$, and the optimal solution in (21) becomes

$$\mathbf{w}_{\bar{n}}^{\mathrm{op}} = \bar{\mathbf{P}}\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}, \tag{24}$$

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where

$$\begin{aligned}\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{op}} &= \underset{\tilde{\mathbf{w}}_{\bar{n}}}{\operatorname{argmin}} \Big\{ \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} - \sigma_{\mathrm{p}} \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} \left(\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{H}} \mathbf{e}_{\bar{n}} + \mathbf{e}_{\bar{n}}^{\mathrm{H}} \tilde{\mathbf{w}}_{\bar{n}}\right) \\ &\quad + \tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{H}} \left(\sigma_{\mathrm{p}}^2 \bar{\mathbf{R}}^{\mathrm{TB}} + \sigma_{\mathrm{z}}^2 \mathbf{I}\right) \tilde{\mathbf{w}}_{\bar{n}} \Big\} \\ &= \left(\sigma_{\mathrm{p}}^2 \bar{\mathbf{R}}^{\mathrm{TB}} + \sigma_{\mathrm{z}}^2 \mathbf{I}\right)^{-1} \mathbf{e}_{\bar{n}} \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} \sigma_{\mathrm{p}}. \end{aligned} \tag{25}$$

The relation in (24) provides an asymptotically optimal estimator with a simple and closed form. Since expression (24) is similar to that of the TB based channel model in (3), we refer to $\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}$ as the TB domain estimator.

Since the TB domain statistical CSI, $\bar{\mathbf{R}}^{\mathrm{TB}}$, is a diagonal matrix, the optimal solution (25) satisfies

$$[\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}]_{\bar{n}'} = 0, \text{ for } \bar{n}' \neq \bar{n}. \tag{26}$$

Therefore, the elements of $\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}$ are non-zero only in the position corresponding to the $\bar{n}$-th non-zero TB, for sufficiently large $M, N_{\mathrm{v}}, N_{\mathrm{F}}$. Thus we can focus on the non-zero element of $\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}$. In this case, the optimal solution in (24) can be rewritten as

$$\mathbf{w}_{\bar{n}}^{\mathrm{op}} = \bar{\mathbf{P}} \mathbf{e}_{\bar{n}} \bar{w}_{\bar{n}}^{\mathrm{op}}, \tag{27}$$

where

$$\begin{aligned}\bar{w}_{\bar{n}}^{\mathrm{op}} &= \underset{\bar{w}_{\bar{n}}}{\operatorname{argmin}} \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} - \sigma_{\mathrm{p}} \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} \left(\bar{w}_{\bar{n}}^{*} + \bar{w}_{\bar{n}}\right) \\ &\quad + \bar{w}_{\bar{n}}^{*} \left(\sigma_{\mathrm{p}}^2 \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} + \sigma_{\mathrm{z}}^2\right) \bar{w}_{\bar{n}} \\ &= \left(\sigma_{\mathrm{p}}^2 \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} + \sigma_{\mathrm{z}}^2\right)^{-1} \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} \sigma_{\mathrm{p}}, \end{aligned} \tag{28}$$

and $\bar{w}_{\bar{n}}^{\mathrm{op}}$ is the low-dimensional TB domain estimator corresponding to the non-zero elements of $\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}$.

Note that the above asymptotically optimal design of channel estimator requires that $M, N_{\mathrm{v}}, N_{\mathrm{F}}$ tend to be infinite. However, in general case, i.e., practical system setup, $M, N_{\mathrm{v}}, N_{\mathrm{F}}$ are always finite, which means that the design of TB domain estimator with only the TB corresponding to $\left[\bar{\mathbf{h}}^{\mathrm{TB}}\right]_{\bar{n}}$ may be not sufficient to obtain a satisfactory performance. In this case, more TBs need to be considered to improve the channel estimation performance. We define the ordered index set of the selected TBs for the estimator design of $\left[\bar{\mathbf{h}}^{\mathrm{TB}}\right]_{\bar{n}}$ as $\hat{\mathcal{N}}_{\bar{n}}^{\mathrm{TB}}$ with $\hat{N}_{\mathrm{TB},\bar{n}} = |\hat{\mathcal{N}}_{\bar{n}}^{\mathrm{TB}}|$ and the corresponding TB selection matrix $\hat{\mathbf{B}}_{\bar{n}} \in \mathbb{C}^{N_{\mathrm{an}} \bar{N}_{\mathrm{de}} N_{\mathrm{do}} \times \hat{N}_{\mathrm{TB},\bar{n}}}$ is

$$\left[\hat{\mathbf{B}}_{\bar{n}}\right]_{i,j} = \begin{cases} 1, \, i = \left[\hat{\mathcal{N}}_{\bar{n}}^{\mathrm{TB}}\right]_j, \\ 0, \, \text{otherwise}. \end{cases} \tag{29}$$

Motivated by (24) and (27), we constrain the estimator of $\left[\bar{\mathbf{h}}^{\mathrm{TB}}\right]_{\bar{n}}$ to have the beam structure as follows

$$\mathbf{w}_{\bar{n}} = \hat{\mathbf{P}} \hat{\mathbf{B}}_{\bar{n}} \hat{\mathbf{w}}_{\bar{n}}, \tag{30}$$

and corresponding MSE is

$$\begin{aligned}&\mathrm{MSE}\left(\hat{\mathbf{w}}_{\bar{n}} \left(\hat{\mathcal{N}}_{\bar{n}}^{\mathrm{TB}}\right)\right) \\ &= \mathbb{E}\left\{ \left\| \left[\bar{\mathbf{h}}^{\mathrm{TB}}\right]_{\bar{n}} - \hat{\mathbf{w}}_{\bar{n}}^{\mathrm{H}} \hat{\mathbf{B}}_{\bar{n}}^{\mathrm{H}} \hat{\mathbf{P}}^{\mathrm{H}} \bar{\mathbf{y}} \right\|^2 \right\} \\ &= \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} - \sigma_{\mathrm{p}} \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} \hat{\mathbf{w}}_{\bar{n}}^{\mathrm{H}} \mathbf{Q}_{\bar{n}} \mathbf{e}_{\bar{n}} - \sigma_{\mathrm{p}} \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} \mathbf{e}_{\bar{n}}^{\mathrm{H}} \mathbf{Q}_{\bar{n}}^{\mathrm{H}} \hat{\mathbf{w}}_{\bar{n}} \\ &\quad + \hat{\mathbf{w}}_{\bar{n}}^{\mathrm{H}} \left(\sigma_{\mathrm{p}}^2 \mathbf{Q}_{\bar{n}} \bar{\mathbf{R}}^{\mathrm{TB}} \mathbf{Q}_{\bar{n}}^{\mathrm{H}} + \sigma_{\mathrm{z}}^2 \hat{\mathbf{B}}_{\bar{n}}^{\mathrm{H}} \hat{\mathbf{P}}^{\mathrm{H}} \hat{\mathbf{P}} \hat{\mathbf{B}}_{\bar{n}}\right) \hat{\mathbf{w}}_{\bar{n}}, \end{aligned} \tag{31}$$

where $\mathbf{Q}_{\bar{n}} \triangleq \hat{\mathbf{B}}_{\bar{n}}^{\mathrm{H}} \hat{\mathbf{P}}^{\mathrm{H}} \bar{\mathbf{P}}$. Then we turn to optimize $\hat{\mathbf{w}}_{\bar{n}}$, whose optimal solution in terms of MSE minimization is

$$\begin{aligned}\hat{\mathbf{w}}_{\bar{n}}^{\mathrm{op}} &= \underset{\hat{\mathbf{w}}_{\bar{n}}}{\operatorname{argmin}} \ \mathrm{MSE}\left(\hat{\mathbf{w}}_{\bar{n}} \left(\hat{\mathcal{N}}_{\bar{n}}^{\mathrm{TB}}\right)\right) \\ &= \left(\sigma_{\mathrm{p}}^2 \mathbf{Q}_{\bar{n}} \bar{\mathbf{R}}^{\mathrm{TB}} \mathbf{Q}_{\bar{n}}^{\mathrm{H}} + \sigma_{\mathrm{z}}^2 \hat{\mathbf{B}}_{\bar{n}}^{\mathrm{H}} \hat{\mathbf{P}}^{\mathrm{H}} \hat{\mathbf{P}} \hat{\mathbf{B}}_{\bar{n}}\right)^{-1} \mathbf{Q}_{\bar{n}} \mathbf{e}_{\bar{n}} \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} \sigma_{\mathrm{p}}. \end{aligned} \tag{32}$$

Thus the resulting SFT domain estimator is $\mathbf{w}_{\bar{n}} = \hat{\mathbf{P}} \hat{\mathbf{B}}_{\bar{n}} \hat{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}$ and the estimate of $\left[\bar{\mathbf{h}}^{\mathrm{TB}}\right]_{\bar{n}}$ is then given by

$$\left[\bar{\mathbf{h}}_{\mathrm{BSCE}}^{\mathrm{TB}}\right]_{\bar{n}} = \mathbf{w}_{\bar{n}}^{\mathrm{H}} \bar{\mathbf{y}} = \left(\hat{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}\right)^{\mathrm{H}} \hat{\mathbf{B}}_{\bar{n}}^{\mathrm{H}} \hat{\mathbf{P}}^{\mathrm{H}} \bar{\mathbf{y}}. \tag{33}$$

Since the SFT domain estimator $\mathbf{w}_{\bar{n}}$ is beam structured with a designed TB domain estimator $\hat{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}$, such a channel estimation approach is referred to as BSCE.

Moreover, for two given TB index sets $\hat{\mathcal{N}}_{\bar{n},1}^{\mathrm{TB}}$ and $\hat{\mathcal{N}}_{\bar{n},2}^{\mathrm{TB}}$ that satisfy $\hat{\mathcal{N}}_{\bar{n},1}^{\mathrm{TB}} \subset \hat{\mathcal{N}}_{\bar{n},2}^{\mathrm{TB}}$, we have

$$\mathrm{MSE}\left(\hat{\mathbf{w}}_{\bar{n}}^{\mathrm{op}} \left(\hat{\mathcal{N}}_{\bar{n},1}^{\mathrm{TB}}\right)\right) \geq \mathrm{MSE}\left(\hat{\mathbf{w}}_{\bar{n}}^{\mathrm{op}} \left(\hat{\mathcal{N}}_{\bar{n},2}^{\mathrm{TB}}\right)\right), \tag{34}$$

which can be easily proved since the case of a smaller index set is a special case of a larger index set. Therefore, we can improve the performance of BSCE by selecting more TBs used for the estimator design, which will lead to an increase in the complexity of the TB domain estimator $\hat{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}$ in (32) as the dimension of the matrix inversion is the number of selected TBs. Therefore, we should select TBs carefully. We next discuss the TB selection for BSCE.

Note that operation $\hat{\mathbf{P}}^{\mathrm{H}} \bar{\mathbf{y}}$ in $\mathrm{MSE}\left(\hat{\mathbf{w}}_{\bar{n}} \left(\hat{\mathcal{N}}_{\bar{n}}^{\mathrm{TB}}\right)\right)$ in (31) can be rewritten as

$$\begin{aligned}\hat{\mathbf{P}}^{\mathrm{H}} \bar{\mathbf{y}} &= \sigma_{\mathrm{p}} \hat{\mathbf{P}}^{\mathrm{H}} \bar{\mathbf{P}} \mathbf{e}_{\bar{n}} \left[\bar{\mathbf{h}}^{\mathrm{TB}}\right]_{\bar{n}} + \sum_{\bar{n}'=0, \bar{n}' \neq \bar{n}}^{\bar{N}_{\mathrm{TB}}-1} \sigma_{\mathrm{p}} \hat{\mathbf{P}}^{\mathrm{H}} \bar{\mathbf{P}} \mathbf{e}_{\bar{n}'} \left[\bar{\mathbf{h}}^{\mathrm{TB}}\right]_{\bar{n}'} \\ &\quad + \hat{\mathbf{P}}^{\mathrm{H}} \mathbf{X}_{\mathrm{c}}^{\mathrm{H}} \mathbf{z}, \end{aligned} \tag{35}$$

which can be seen as a transformation of the received signal from SFT domain to TB domain, then operation $\hat{\mathbf{B}}_{\bar{n}}^{\mathrm{H}} \hat{\mathbf{P}}^{\mathrm{H}} \bar{\mathbf{y}}$ is to select the corresponding part of the TB domain received signal according to TB index set $\hat{\mathcal{N}}_{\bar{n}}^{\mathrm{TB}}$ to design TB domain estimator $\hat{\mathbf{w}}_{\bar{n}}$, which is

$$\begin{aligned}\hat{\mathbf{B}}_{\bar{n}}^{\mathrm{H}} \hat{\mathbf{P}}^{\mathrm{H}} \bar{\mathbf{y}} &= \sigma_{\mathrm{p}} \mathbf{Q}_{\bar{n}} \mathbf{e}_{\bar{n}} \left[\bar{\mathbf{h}}^{\mathrm{TB}}\right]_{\bar{n}} + \sum_{\bar{n}'=0, \bar{n}' \neq \bar{n}}^{\bar{N}_{\mathrm{TB}}-1} \sigma_{\mathrm{p}} \mathbf{Q}_{\bar{n}} \mathbf{e}_{\bar{n}'} \left[\bar{\mathbf{h}}^{\mathrm{TB}}\right]_{\bar{n}'} \\ &\quad + \hat{\mathbf{B}}_{\bar{n}}^{\mathrm{H}} \hat{\mathbf{P}}^{\mathrm{H}} \mathbf{X}_{\mathrm{c}}^{\mathrm{H}} \mathbf{z}. \end{aligned} \tag{36}$$

However, due to the finite $M, N_{\mathrm{v}}, N_{\mathrm{F}}$, the columns of $\hat{\mathbf{P}}$ become non-orthogonal, i.e.,

$$\begin{aligned}&\left[\hat{\mathbf{P}}^{\mathrm{H}} \hat{\mathbf{P}}\right]_{i,j} \\ &= \Upsilon_M\left(\frac{2 f_{\mathrm{c}} \Delta_\tau (n_{\mathrm{an}} - n'_{\mathrm{an}})}{N_{\mathrm{an}}}\right) \Upsilon_{N_{\mathrm{v}}}\left(\frac{N_\tau (n_{\mathrm{de}} - n'_{\mathrm{de}})}{N_{\mathrm{de}} N_{\mathrm{v}}}\right) \Upsilon_{N_{\mathrm{F}}}\left(\frac{N_{\mathrm{d}} (n'_{\mathrm{do}} - n_{\mathrm{do}})}{N_{\mathrm{do}} N_{\mathrm{F}}}\right), \end{aligned} \tag{37}$$

where $\Upsilon_N(x) = e^{\jmath \pi x (N-1)} \sin(\pi N x) / (N \sin(\pi x))$, $n_{\mathrm{do}} = \lfloor i / (\bar{N}_{\mathrm{de}} N_{\mathrm{an}}) \rfloor$, $n_{\mathrm{de}} = \langle \lfloor i/(N_{\mathrm{an}}) \rfloor \rangle_{\bar{N}_{\mathrm{de}}}$, $n_{\mathrm{an}} = \langle i \rangle_{N_{\mathrm{an}}}$, $n'_{\mathrm{do}} = \lfloor j / (\bar{N}_{\mathrm{de}} N_{\mathrm{an}}) \rfloor$, $n'_{\mathrm{de}} = \langle \lfloor j/(N_{\mathrm{an}}) \rfloor \rangle_{\bar{N}_{\mathrm{de}}}$, $n'_{\mathrm{an}} = \langle j \rangle_{N_{\mathrm{an}}}$, and $\langle \cdot \rangle_N$ denotes the modulo-$N$ operation. An example of $\hat{\mathbf{P}}^{\mathrm{H}} \hat{\mathbf{P}}$ is shown in Fig. 5. This makes $\hat{\mathbf{P}}^{\mathrm{H}} \bar{\mathbf{P}} \mathbf{e}_{\bar{n}} \left[\bar{\mathbf{h}}^{\mathrm{TB}}\right]_{\bar{n}}$ non-zero not only on the TB corresponding to $\left[\bar{\mathbf{h}}^{\mathrm{TB}}\right]_{\bar{n}}$, but also

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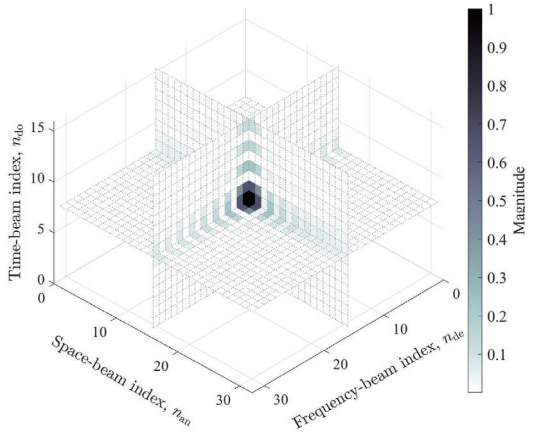



Fig. 5. Slices of the magnitude of $\left[\hat{\mathbf{P}}^{\rm H}\hat{\mathbf{P}}\right]_{:,j}$, where $M = N_{\rm v} = 16$, $N_{\rm S} = 14$, $N_{\rm F} = 8$, $N_\tau = 4$, $N_{\rm d} = 8$, $N_{\rm an} = 32$, $N_{\rm de} = 8$, $N_{\rm do} = 16$, $n'_{\rm an} = \frac{N_{\rm an}}{2} - 1$, $n'_{\rm de} = \frac{N_{\rm de}}{2} - 1$, $n'_{\rm do} = \frac{N_{\rm do}}{2} - 1$.

on its surrounding TBs, which is also referred to as spectral leakage. Such a spectral leakage will cause the TB domain channel elements located in the vicinity of $\left[\bar{\mathbf{h}}^{\rm TB}\right]_{\bar{n}}$ to interfere with $\left[\bar{\mathbf{h}}^{\rm TB}\right]_{\bar{n}}$, resulting in inter-TB interference as shown in the second term in the right-hand side of (36). Moreover, the interference caused by TB domain channel elements closer to $\left[\bar{\mathbf{h}}^{\rm TB}\right]_{\bar{n}}$ is generally greater. To take into account the spectral leakage and the interference caused by surrounding TBs when designing the TB domain channel estimator for $\left[\bar{\mathbf{h}}^{\rm TB}\right]_{\bar{n}}$, we can select the TBs in the vicinity of $\left[\bar{\mathbf{h}}^{\rm TB}\right]_{\bar{n}}$ besides the TB occupied by itself. Therefore, motivated by Fig. 5, when we need to add more TBs in addition to the TB that $\left[\bar{\mathbf{h}}^{\rm TB}\right]_{\bar{n}}$ occupies to design the TB domain estimator, we can select the TBs in the vicinity of the TB that $\left[\bar{\mathbf{h}}^{\rm TB}\right]_{\bar{n}}$ occupies in a cross pattern as shown on the left of Fig. 6. We can also select more TBs in a cubic pattern as shown on the right of Fig. 6 to achieve better performance. Note that selecting more TBs for BSCE will increase its complexity since the dimension of the designed TB domain channel estimator is the number of selected TBs.

### C. BSCE With Window Function

Since the inter-TB interference is caused by the non-orthogonality among TBs as shown in the second term in the right-hand side of (36) and term $\sigma_{\rm p}^2\mathbf{Q}_{\bar{n}}\bar{\mathbf{R}}^{\rm TB}\mathbf{Q}_{\bar{n}}^{\rm H}$ in (32), we introduce a window function to reduce the spectral leakage and make the energy of each TB domain channel element $\left[\bar{\mathbf{h}}^{\rm TB}\right]_{\bar{n}}$ more concentrated. Define $\mathbf{T} = {\rm diag}\left\{\mathbf{t}_{\rm an}\otimes\mathbf{t}_{\rm de}\otimes\mathbf{t}_{\rm do}\right\}$, where $\mathbf{t}_{\rm an}\in\mathbb{C}^M$, $\mathbf{t}_{\rm de}\in\mathbb{C}^{N_{\rm v}}$ and $\mathbf{t}_{\rm do}\in\mathbb{C}^{N_{\rm F}}$ are window functions. Similar to (31)-(33), we constrain the estimator of $\left[\bar{\mathbf{h}}^{\rm TB}\right]_{\bar{n}}$ to have the structure $\mathbf{w}_{\bar{n}}=\hat{\mathbf{P}}\hat{\mathbf{B}}_{\bar{n}}\hat{\mathbf{w}}_{\bar{n}}^{\rm win}$ and apply it to the received signal after windowing (i.e., $\mathbf{T}^{\rm H}\bar{\mathbf{y}}$). Then the corresponding MSE is

$$\mathrm{MSE}\left(\hat{\mathbf{w}}_{\bar{n}}^{\rm win}\left(\hat{\mathcal{N}}_{\bar{n}}^{\rm TB}\right)\right)=\mathbb{E}\left\{\left\|\left[\bar{\mathbf{h}}^{\rm TB}\right]_{\bar{n}}-\left(\hat{\mathbf{w}}_{\bar{n}}^{\rm win}\right)^{\rm H}\hat{\mathbf{B}}_{\bar{n}}^{\rm H}\hat{\mathbf{P}}^{\rm H}\mathbf{T}^{\rm H}\bar{\mathbf{y}}\right\|^2\right\}, \tag{38}$$

and optimal $\hat{\mathbf{w}}_{\bar{n}}^{\rm win}$ in terms of MSE minimization is

$$\begin{aligned}\hat{\mathbf{w}}_{\bar{n}}^{\rm win,op}&=\underset{\hat{\mathbf{w}}_{\bar{n}}^{\rm win}}{\rm argmin}\,\mathrm{MSE}\left(\hat{\mathbf{w}}_{\bar{n}}^{\rm win}\left(\hat{\mathcal{N}}_{\bar{n}}^{\rm TB}\right)\right)\\&=\left(\sigma_{\rm p}^2\hat{\mathbf{Q}}_{\bar{n}}\bar{\mathbf{R}}^{\rm TB}\hat{\mathbf{Q}}_{\bar{n}}^{\rm H}+\sigma_{\rm z}^2\hat{\mathbf{B}}_{\bar{n}}^{\rm H}\hat{\mathbf{P}}^{\rm H}\mathbf{T}^{\rm H}\mathbf{T}\hat{\mathbf{P}}\hat{\mathbf{B}}_{\bar{n}}\right)^{-1}\hat{\mathbf{Q}}_{\bar{n}}\mathbf{e}_{\bar{n}}\left[\bar{\mathbf{R}}^{\rm TB}\right]_{\bar{n},\bar{n}}\sigma_{\rm p},\end{aligned} \tag{39}$$

where $\hat{\mathbf{Q}}_{\bar{n}}\triangleq\hat{\mathbf{B}}_{\bar{n}}^{\rm H}\hat{\mathbf{P}}^{\rm H}\mathbf{T}^{\rm H}\bar{\mathbf{P}}$ and estimate of $\left[\bar{\mathbf{h}}^{\rm TB}\right]_{\bar{n}}$ is

$$\left[\bar{\mathbf{h}}_{\rm BSCE,win}^{\rm TB}\right]_{\bar{n}}=\left(\hat{\mathbf{w}}_{\bar{n}}^{\rm win,op}\right)^{\rm H}\hat{\mathbf{B}}_{\bar{n}}^{\rm H}\hat{\mathbf{P}}^{\rm H}\mathbf{T}^{\rm H}\bar{\mathbf{y}}. \tag{40}$$

Comparing (39) with (32), the term $\hat{\mathbf{P}}^{\rm H}\bar{\mathbf{P}}$ in $\mathbf{Q}_{\bar{n}}$ in (32) becomes $\hat{\mathbf{P}}^{\rm H}\mathbf{T}^{\rm H}\bar{\mathbf{P}}$ in $\hat{\mathbf{Q}}_{\bar{n}}$ in (39). When the window functions, for example, are selected as cosine windows, i.e., $\left[\mathbf{t}_{\rm an}\right]_i = \cos\left(\left(\frac{i}{M}-\frac{M-1}{2M}\right)\pi\right)$, $\left[\mathbf{t}_{\rm de}\right]_i = \cos\left(\left(\frac{i}{N_{\rm v}}-\frac{N_{\rm v}-1}{2N_{\rm v}}\right)\pi\right)$, and $\left[\mathbf{t}_{\rm do}\right]_i = \cos\left(\left(\frac{i}{N_{\rm F}}-\frac{N_{\rm F}-1}{2N_{\rm F}}\right)\pi\right)$, we have

$$\begin{aligned}&\left[\hat{\mathbf{P}}^{\rm H}\mathbf{T}^{\rm H}\hat{\mathbf{P}}\right]_{i,j}\\&=\tilde{\Upsilon}_M\left(\frac{2f_{\rm c}\Delta_\tau(n_{\rm an}-n'_{\rm an})}{N_{\rm an}}\right)\tilde{\Upsilon}_{N_{\rm v}}\left(\frac{N_\tau(n_{\rm de}-n'_{\rm de})}{N_{\rm de}N_{\rm v}}\right)\tilde{\Upsilon}_{N_{\rm F}}\left(\frac{N_{\rm d}(n'_{\rm do}-n_{\rm do})}{N_{\rm do}N_{\rm F}}\right),\end{aligned} \tag{41}$$

where $\tilde{\Upsilon}_N(x)=\frac{1}{2}\left(e^{-\bar{\jmath}\frac{N-1}{2N}\pi}\Upsilon_N\left(x+\frac{1}{2N}\right)+e^{\bar{\jmath}\frac{N-1}{2N}\pi}\Upsilon_N\left(x-\frac{1}{2N}\right)\right)$, $n_{\rm do}=\left\lfloor i/\left(\bar{N}_{\rm de}N_{\rm an}\right)\right\rfloor$, $n_{\rm de}=\left\langle\left\lfloor i/(N_{\rm an})\right\rfloor\right\rangle_{\bar{N}_{\rm de}}$, $n_{\rm an}=\langle i\rangle_{N_{\rm an}}$, $n'_{\rm do}=\left\lfloor j/\left(\bar{N}_{\rm de}N_{\rm an}\right)\right\rfloor$, $n'_{\rm de}=\left\langle\left\lfloor j/(N_{\rm an})\right\rfloor\right\rangle_{\bar{N}_{\rm de}}$, $n'_{\rm an}=\langle j\rangle_{N_{\rm an}}$. In this case, an example of $\hat{\mathbf{P}}^{\rm H}\mathbf{T}^{\rm H}\hat{\mathbf{P}}$ is shown in Fig. 7. From the figure, the energy is more concentrated compared with Fig. 5 and the spectral leakage is reduced, thus similar performance can be achieved using fewer TBs, which results in a lower complexity. Actually, BSCE in (32)-(33) is a special case of (39)-(40), where the window functions are selected as rectangular windows. BSCE in (39)-(40) can be referred to as a windowed BSCE.

### D. Turbo-BSCE

Although we can obtain a good performance by selecting a large number of TBs when we design the TB domain estimator for each TB domain channel element, it will lead to a high computational complexity since the dimension of the matrix inversion in (32) and (39) is the number of selected TBs. On the other hand, selecting fewer TBs for the TB domain estimator design of each TB domain channel element will reduce complexity but may also result in unsatisfactory performance. To tackle this trade-off between performance and complexity, we introduce turbo principle into BSCE, referred to as turbo-BSCE, to improve its performance, especially when fewer TBs are selected.

The turbo principle has been widely used in channel decoding [46], [47], equalization [48], [49], and multi-user detection [50], [51], where the processed output values of each iteration are used as a priori input for the next iteration to obtain a better performance. Based on the turbo principle, we can extend BSCE into an iterative form and use the output of each iteration as a priori information

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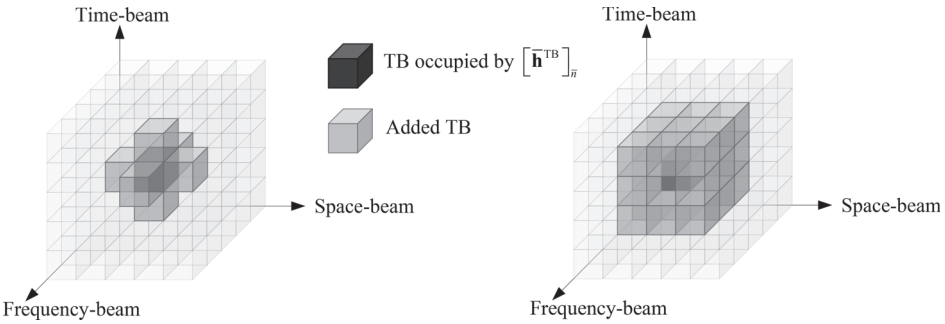



Fig. 6. Illustration of TB selection. Besides the TB occupied by $\left[\bar{\mathbf{h}}^{\mathrm{TB}}\right]_{\bar{n}}$, more TBs can be selected in a cross pattern as shown on the left or a cubic pattern as shown on the right.

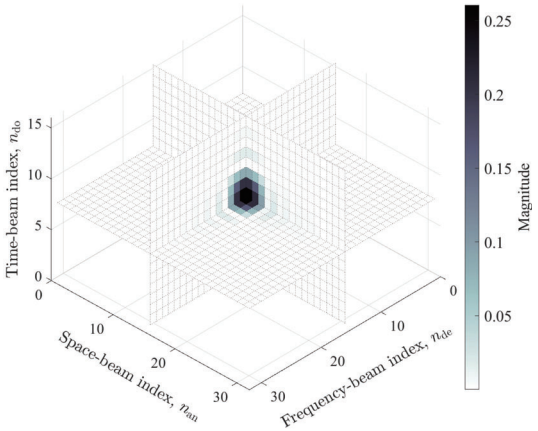



Fig. 7. Slices of the magnitude of $\left[\hat{\mathbf{P}}^{\mathrm{H}}\mathbf{T}^{\mathrm{H}}\hat{\mathbf{P}}\right]_{:,j}$, where $M = N_{\mathrm{v}} = 16$, $N_{\mathrm{S}} = 14$, $N_{\mathrm{F}} = 8$, $N_{\tau} = 4$, $N_{\mathrm{d}} = 8$, $N_{\mathrm{an}} = 32$, $N_{\mathrm{de}} = 8$, $N_{\mathrm{do}} = 16$, $n'_{\mathrm{an}} = \frac{N_{\mathrm{an}}}{2} - 1$, $n'_{\mathrm{de}} = \frac{N_{\mathrm{de}}}{2} - 1$, $n'_{\mathrm{do}} = \frac{N_{\mathrm{do}}}{2} - 1$.

for the input of the next iteration to obtain a new estimate. Since BSCE in (32)-(33) is a special case of that in (39)-(40) with the rectangular window, we derive turbo-BSCE based on this windowed BSCE in the subsequent discussion. Specifically, in the first iteration, the prior mean of each TB domain channel element is zero and the corresponding prior covariance is the TB domain statistical CSI $\bar{\mathbf{R}}^{\mathrm{TB}}$, thus the resulting TB domain estimator and the estimate of the TB domain channel are given by (39) and (40), respectively, which are re-expressed as $\hat{\mathbf{w}}_{\bar{n}}^{\mathrm{turbo},(1)}$ and $\left[\bar{\mathbf{h}}_{\mathrm{turbo}}^{\mathrm{TB},(1)}\right]_{\bar{n}}$. Then a posteriori variance of the estimation result is given by

$$\left[\bar{\mathbf{R}}_{\mathrm{turbo}}^{\mathrm{TB},(1)}\right]_{\bar{n},\bar{n}} = \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} - \left(\hat{\mathbf{w}}_{\bar{n}}^{\mathrm{turbo},(1)}\right)^{\mathrm{H}} \hat{\mathbf{Q}}_{\bar{n}} \mathbf{e}_{\bar{n}} \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} \sigma_{\mathrm{p}}. \tag{42}$$

Next, we take the output estimation result and a posterior variance of each iteration as the input prior mean and variance of each TB domain channel element to the next iteration. Furthermore, for the estimation of each TB domain channel element $\left[\bar{\mathbf{h}}^{\mathrm{TB}}\right]_{\bar{n}}$, we need to exclude the contribution of itself according to the turbo principle [46]. Thus when we perform BSCE of $\left[\bar{\mathbf{h}}^{\mathrm{TB}}\right]_{\bar{n}}$ at the $t$-th iteration, we set the input prior mean and covariance as

$$\mathbb{E}\left\{\bar{\mathbf{h}}^{\mathrm{TB}}\right\} = \bar{\mathbf{h}}_{\mathrm{turbo}}^{\mathrm{TB},(t-1)} - \left[\bar{\mathbf{h}}_{\mathrm{turbo}}^{\mathrm{TB},(t-1)}\right]_{\bar{n}} \mathbf{e}_{\bar{n}}, \tag{43}$$

$$\mathrm{Cov}\left\{\bar{\mathbf{h}}^{\mathrm{TB}}\right\} = \bar{\mathbf{R}}_{\mathrm{turbo}}^{\mathrm{TB},(t-1)} - \left(\left[\bar{\mathbf{R}}_{\mathrm{turbo}}^{\mathrm{TB},(t-1)}\right]_{\bar{n},\bar{n}} - \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}}\right) \mathbf{e}_{\bar{n}} \mathbf{e}_{\bar{n}}^{\mathrm{H}}. \tag{44}$$

**Algorithm 1** Turbo-BSCE

**Input**: $\bar{\mathbf{y}}$, $\mathbf{T}$, $\hat{\mathbf{P}}$, $\bar{\mathbf{R}}^{\mathrm{TB}}$, $\left\{\hat{\mathbf{B}}_{\bar{n}}, \bar{n} = 0,1,\cdots,\bar{N}_{\mathrm{TB}} - 1\right\}$, $\sigma_{\mathrm{p}}^2$, $\sigma_z^2$, iteration number $T$

**Initialize:** $\bar{\mathbf{h}}_{\mathrm{turbo}}^{\mathrm{TB},(0)} = \mathbf{0}$, $\bar{\mathbf{R}}_{\mathrm{turbo}}^{\mathrm{TB},(0)} = \bar{\mathbf{R}}^{\mathrm{TB}}$, $\tilde{\mathbf{y}} = \hat{\mathbf{P}}^{\mathrm{H}}\mathbf{T}^{\mathrm{H}}\bar{\mathbf{y}}$

**for** $t = 1$ *to* $T$ **do**

  $\forall \bar{n}$, calculate $\hat{\mathbf{y}}_{\bar{n}}^{(t)}$ by (47).

  $\forall \bar{n}$, design TB domain estimator $\hat{\mathbf{w}}_{\bar{n}}^{\mathrm{turbo},(t)}$ by (46).

  $\forall \bar{n}$, calculate channel estimate $\left[\bar{\mathbf{h}}_{\mathrm{turbo}}^{\mathrm{TB},(t)}\right]_{\bar{n}}$ by (45).

  $\forall \bar{n}$, calculate $\left[\bar{\mathbf{R}}_{\mathrm{turbo}}^{\mathrm{TB},(t)}\right]_{\bar{n},\bar{n}}$ by (48).

**end**

**Output**: $\bar{\mathbf{h}}_{\mathrm{turbo}}^{\mathrm{TB}} = \bar{\mathbf{h}}_{\mathrm{turbo}}^{\mathrm{TB},(t)}$

Hence, the estimation of $\left[\bar{\mathbf{h}}^{\mathrm{TB}}\right]_{\bar{n}}$ at the $t$-th iteration is

$$\left[\bar{\mathbf{h}}_{\mathrm{turbo}}^{\mathrm{TB},(t)}\right]_{\bar{n}} = \left(\hat{\mathbf{w}}_{\bar{n}}^{\mathrm{turbo},(t)}\right)^{\mathrm{H}} \hat{\mathbf{y}}_{\bar{n}}^{(t)}, \tag{45}$$

where

$$\hat{\mathbf{w}}_{\bar{n}}^{\mathrm{turbo},(t)} = \left(\sigma_{\mathrm{p}}^2 \hat{\mathbf{Q}}_{\bar{n}} \mathrm{Cov}\left\{\bar{\mathbf{h}}^{\mathrm{TB}}\right\} \hat{\mathbf{Q}}_{\bar{n}}^{\mathrm{H}} + \sigma_z^2 \hat{\mathbf{B}}_{\bar{n}}^{\mathrm{H}} \hat{\mathbf{P}}^{\mathrm{H}} \mathbf{T}^{\mathrm{H}} \mathbf{T} \hat{\mathbf{P}} \hat{\mathbf{B}}_{\bar{n}}\right)^{-1} \times \hat{\mathbf{Q}}_{\bar{n}} \mathbf{e}_{\bar{n}} \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} \sigma_{\mathrm{p}}, \tag{46}$$

and

$$\hat{\mathbf{y}}_{\bar{n}}^{(t)} = \hat{\mathbf{B}}_{\bar{n}}^{\mathrm{H}} \tilde{\mathbf{y}} - \sigma_{\mathrm{p}} \hat{\mathbf{Q}}_{\bar{n}} \mathbb{E}\left\{\bar{\mathbf{h}}^{\mathrm{TB}}\right\}, \tag{47}$$

where $\tilde{\mathbf{y}} \triangleq \hat{\mathbf{P}}^{\mathrm{H}}\mathbf{T}^{\mathrm{H}}\bar{\mathbf{y}}$. Moreover, a posterior variance of the $t$-th iteration is

$$\left[\bar{\mathbf{R}}_{\mathrm{turbo}}^{\mathrm{TB},(t)}\right]_{\bar{n},\bar{n}} = \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} - \left(\hat{\mathbf{w}}_{\bar{n}}^{\mathrm{turbo},(t)}\right)^{\mathrm{H}} \hat{\mathbf{Q}}_{\bar{n}} \mathbf{e}_{\bar{n}} \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} \sigma_{\mathrm{p}}. \tag{48}$$

This process repeats iteratively until convergence or a predetermined iteration number $T$ has been reached. The propsoed turbo-BSCE algorithm is summarized in Algorithm 1 and illustrated in Fig. 8.

## IV. Low-Complexity Design and Implementation

In this section, we present a low-complexity design and implementation for BSCE by exploiting the characteristics of the TB matrix. Moreover, we show that the computation of the TB domain estimator involves only real-valued matrices.

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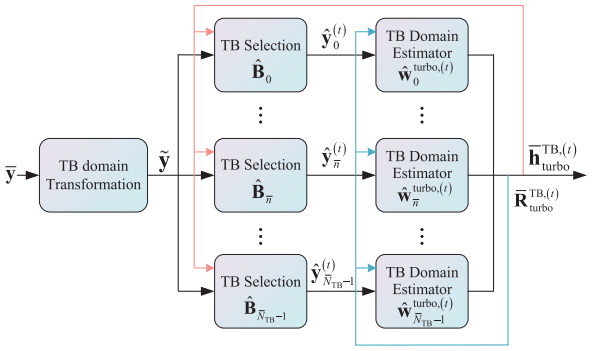



Fig. 8. Illustration of turbo-BSCE.

### A. A Low-Complexity Design

From Algorithm 1, BSCE mainly consists of two parts: design the TB domain estimator (step 4) and calculate the channel estimate with the designed TB domain estimator (step 5). In this subsection, we first present a low-complexity design of the TB domain estimator.

The design complexity of BSCE mainly comes from step 4 in Algorithm 1, i.e., the design of the TB domain estimator in (46). We first show that the computation of the TB domain estimator in (46) involves only real-valued matrices. Recalling the definition of $\hat{\mathbf{P}}$ in (14), it can be checked that the columns of matrices $\mathbf{V}$, $\bar{\mathbf{U}}$, $\mathbf{D}$ exhibit conjugate centrosymmetry, i.e., $\tilde{\mathbf{I}}_M\mathbf{V}=\mathbf{V}^*$, $\tilde{\mathbf{I}}_{N_\mathrm{v}}\bar{\mathbf{U}}=\bar{\mathbf{U}}^*$, and $\tilde{\mathbf{I}}_{N_\mathrm{F}}\mathbf{D}=\mathbf{D}^*$, where $\tilde{\mathbf{I}}_N\in\mathbb{C}^{N\times N}$ is the reverse permutation matrix and satisfies $\tilde{\mathbf{I}}_N\tilde{\mathbf{I}}_N=\mathbf{I}_N$. Therefore, when the selected window functions satisfy the conjugate centrosymmetry, we have $\tilde{\mathbf{I}}_M\mathrm{diag}\{\mathbf{t}_\mathrm{an}\}\mathbf{V}=\mathrm{diag}\{\mathbf{t}_\mathrm{an}^*\}\mathbf{V}^*$, $\tilde{\mathbf{I}}_{N_\mathrm{v}}\mathrm{diag}\{\mathbf{t}_\mathrm{de}\}\bar{\mathbf{U}}=\mathrm{diag}\{\mathbf{t}_\mathrm{de}^*\}\bar{\mathbf{U}}^*$, $\tilde{\mathbf{I}}_{N_\mathrm{F}}\mathrm{diag}\{\mathbf{t}_\mathrm{do}\}\mathbf{D}=\mathrm{diag}\{\mathbf{t}_\mathrm{do}^*\}\mathbf{D}^*$, and

$$\begin{aligned}\hat{\mathbf{P}}^\mathrm{H}\mathbf{T}^\mathrm{H}\hat{\mathbf{P}}&=\hat{\mathbf{P}}^\mathrm{H}\mathbf{T}^\mathrm{H}\left(\tilde{\mathbf{I}}_{N_\mathrm{F}}\otimes\tilde{\mathbf{I}}_{N_\mathrm{v}}\otimes\tilde{\mathbf{I}}_{N_\mathrm{an}}\right)\left(\tilde{\mathbf{I}}_{N_\mathrm{F}}\otimes\tilde{\mathbf{I}}_{N_\mathrm{v}}\otimes\tilde{\mathbf{I}}_{N_\mathrm{an}}\right)\hat{\mathbf{P}}\\&=\hat{\mathbf{P}}^\mathrm{T}\mathbf{T}^\mathrm{T}\hat{\mathbf{P}}^*,\end{aligned}\tag{49}$$

which means that the matrix $\hat{\mathbf{P}}^\mathrm{H}\mathbf{T}^\mathrm{H}\hat{\mathbf{P}}$ is a real-valued matrix. Similarly, it can be checked that $\hat{\mathbf{P}}^\mathrm{H}\mathbf{T}^\mathrm{H}\mathbf{T}\hat{\mathbf{P}}$ is also a real-valued matrix. Thus the computation of the TB domain estimator $\hat{\mathbf{w}}_{\bar{n}}^{\mathrm{turbo},(t)}$ in (46) involves only real-valued matrices, thus reducing the computational complexity.

Next, since $\hat{\mathbf{P}}^\mathrm{H}\mathbf{T}^\mathrm{H}\hat{\mathbf{P}}$ and $\hat{\mathbf{P}}^\mathrm{H}\mathbf{T}^\mathrm{H}\mathbf{T}\hat{\mathbf{P}}$ involved in (46) (note that $\bar{\mathbf{P}}=\left[\hat{\mathbf{P}}\right]_{:,\bar{N}^\mathrm{TB}}$) are not related to the received signal and the specific UT, we can pre-calculate them offline and store them, which will lead to a computational complexity $\mathcal{O}\left(\left(N_\mathrm{an}\bar{N}_\mathrm{de}N_\mathrm{do}\right)^2MN_\mathrm{v}N_\mathrm{F}\right)$ and a storage complexity $\mathcal{O}\left(\left(N_\mathrm{an}\bar{N}_\mathrm{de}N_\mathrm{do}\right)^2\right)$ and may be large in practical systems. Here, we present a method to compute and store with low complexity by exploiting the structure of the TB matrix.

We take the computation of $\hat{\mathbf{P}}^\mathrm{H}\mathbf{T}^\mathrm{H}\hat{\mathbf{P}}$ as an example, and the computation of $\hat{\mathbf{P}}^\mathrm{H}\mathbf{T}^\mathrm{H}\mathbf{T}\hat{\mathbf{P}}$ can be treated in a similar way. Note that $\hat{\mathbf{P}}^\mathrm{H}\mathbf{T}^\mathrm{H}\hat{\mathbf{P}}$ can be rewritten as

$$\begin{aligned}\hat{\mathbf{P}}^\mathrm{H}\mathbf{T}^\mathrm{H}\hat{\mathbf{P}}=&\underbrace{\left([\mathbf{D}]_{\mathcal{P},:}^\mathrm{H}\,\mathrm{diag}\{\mathbf{t}_\mathrm{do}^*\}[\mathbf{D}]_{\mathcal{P},:}\right)}_{\mathbf{\Phi}_\mathrm{do}}\\&\otimes\underbrace{\left(\bar{\mathbf{U}}^\mathrm{H}\mathrm{diag}\{\mathbf{t}_\mathrm{de}^*\}\bar{\mathbf{U}}\right)}_{\mathbf{\Phi}_\mathrm{de}}\otimes\underbrace{\left(\mathbf{V}^\mathrm{H}\mathrm{diag}\{\mathbf{t}_\mathrm{an}^*\}\mathbf{V}\right)}_{\mathbf{\Phi}_\mathrm{an}}.\end{aligned}\tag{50}$$

Then we have

$$\left[\hat{\mathbf{P}}^\mathrm{H}\mathbf{T}^\mathrm{H}\hat{\mathbf{P}}\right]_{i,j}=[\mathbf{\Phi}_\mathrm{do}]_{n_\mathrm{do},n'_\mathrm{do}}[\mathbf{\Phi}_\mathrm{de}]_{n_\mathrm{de},n'_\mathrm{de}}[\mathbf{\Phi}_\mathrm{an}]_{n_\mathrm{an},n'_\mathrm{an}},\tag{51}$$

where

$$\begin{aligned}&n_\mathrm{do}=\left\lfloor i/\left(\bar{N}_\mathrm{de}N_\mathrm{an}\right)\right\rfloor,n_\mathrm{de}=\langle\lfloor i/(N_\mathrm{an})\rfloor\rangle_{\bar{N}_\mathrm{de}},n_\mathrm{an}=\langle i\rangle_{N_\mathrm{an}},\\&n'_\mathrm{do}=\left\lfloor j/\left(\bar{N}_\mathrm{de}N_\mathrm{an}\right)\right\rfloor,n'_\mathrm{de}=\langle\lfloor j/(N_\mathrm{an})\rfloor\rangle_{\bar{N}_\mathrm{de}},n'_\mathrm{an}=\langle j\rangle_{N_\mathrm{an}}.\end{aligned}$$

Moreover, it can be checked that

$$[\mathbf{\Phi}_\mathrm{an}]_{n_\mathrm{an}:N_\mathrm{an}-1,n_\mathrm{an}}=[\mathbf{\Phi}_\mathrm{an}]_{0:N_\mathrm{an}-1-n_\mathrm{an},0},\tag{52}$$

where $n_\mathrm{an}\in\{0,\cdots,N_\mathrm{an}-1\}$ and $\mathbf{\Phi}_\mathrm{an}$ is a Hermitian matrix. Thus $\mathbf{\Phi}_\mathrm{an}$ can be constructued by its first column $[\mathbf{\Phi}_\mathrm{an}]_{:,0}=\mathbf{V}^\mathrm{H}\mathrm{diag}\{\mathbf{t}_\mathrm{an}^*\}[\mathbf{V}]_{:,0}$, i.e.,

$$[\mathbf{\Phi}_\mathrm{an}]_{n_\mathrm{an},n'_\mathrm{an}}=\begin{cases}[\mathbf{\Phi}_\mathrm{an}]_{n_\mathrm{an}-n'_\mathrm{an},0},&n_\mathrm{an}\geq n'_\mathrm{an},\\[\mathbf{\Phi}_\mathrm{an}]^*_{n'_\mathrm{an}-n_\mathrm{an},0},&n_\mathrm{an}<n'_\mathrm{an}.\end{cases}\tag{53}$$

Similarly, $\mathbf{\Phi}_\mathrm{de}$ and $\mathbf{\Phi}_\mathrm{do}$ can also be constructued by their first columns $[\mathbf{\Phi}_\mathrm{de}]_{:,0}=\bar{\mathbf{U}}^\mathrm{H}\mathrm{diag}\{\mathbf{t}_\mathrm{de}^*\}\left[\bar{\mathbf{U}}\right]_{:,0}$ and $[\mathbf{\Phi}_\mathrm{do}]_{:,0}=[\mathbf{D}]_{\mathcal{P},:}^\mathrm{H}\mathrm{diag}\{\mathbf{t}_\mathrm{do}^*\}[\mathbf{D}]_{\mathcal{P},0}$. Denote $\boldsymbol{\omega}_\mathrm{an}=\frac{1}{\sqrt{M}}\left[\omega_0^\mathrm{an},\cdots,\omega_{N_\mathrm{an}-1}^\mathrm{an}\right]^\mathrm{T}$, $\boldsymbol{\omega}_\mathrm{de}=\frac{1}{\sqrt{N_\mathrm{v}}}\left[\omega_0^\mathrm{do},\cdots,\omega_{\bar{N}_\mathrm{de}-1}^\mathrm{do}\right]^\mathrm{T}$, and $\boldsymbol{\omega}_\mathrm{do}=\frac{1}{\sqrt{N_\mathrm{F}}}\left[\omega_0^\mathrm{do},\cdots,\omega_{N_\mathrm{do}-1}^\mathrm{do}\right]^\mathrm{T}$. Note that $\bar{\mathbf{U}}=\left[\mathbf{F}_{\bar{N}_\mathrm{de}}\right]_{0:N_\mathrm{v}-1,:}\mathrm{diag}\{\boldsymbol{\omega}_\mathrm{de}\}$ is a partial DFT matrix, where $\mathbf{F}_N$ denotes the $N$-dimensional unitary DFT matrix. Moreover, $\mathbf{V}$ and $[\mathbf{D}]_{\mathcal{P},:}$ can be regarded as chirp z-transform (CZT) matrices, which can be rewritten as

$$\begin{aligned}\mathbf{V}=&\,\mathrm{diag}\left\{\boldsymbol{\xi}_{W_\mathrm{S},N_\mathrm{an},0,M}\right\}\mathbf{F}_{N_{(\mathrm{S})}\times M}^\mathrm{H}\\&\times\mathrm{diag}\left\{\tilde{\boldsymbol{\xi}}_{W_\mathrm{S},M,N_\mathrm{an},N_{(\mathrm{S})}}\right\}\mathbf{F}_{N_{(\mathrm{S})}\times N_\mathrm{an}}\\&\times\mathrm{diag}\left\{\boldsymbol{\xi}_{W_\mathrm{S},0,0,N_\mathrm{an}}\right\}\mathrm{diag}\left\{\boldsymbol{\omega}_\mathrm{an}\right\},\end{aligned}\tag{54a}$$

$$\begin{aligned}[\mathbf{D}]_{\mathcal{P},:}=&\,\mathrm{diag}\left\{\boldsymbol{\xi}_{W_\mathrm{T},N_\mathrm{do},\frac{n_\mathrm{p}N_\mathrm{do}}{N_\mathrm{S}},N_\mathrm{F}}\right\}\mathbf{F}_{N_{(\mathrm{T})}\times N_\mathrm{F}}^\mathrm{H}\\&\times\mathrm{diag}\left\{\tilde{\boldsymbol{\xi}}_{W_\mathrm{T},N_\mathrm{F},N_\mathrm{do},N_{(\mathrm{T})}}\right\}\mathbf{F}_{N_{(\mathrm{T})}\times N_\mathrm{do}}\\&\times\mathrm{diag}\left\{\boldsymbol{\xi}_{W_\mathrm{T},-\frac{2n_\mathrm{p}}{N_\mathrm{S}},0,N_\mathrm{do}}\right\}\mathrm{diag}\left\{\boldsymbol{\omega}_\mathrm{do}\right\},\end{aligned}\tag{54b}$$

respectively, where $W_\mathrm{S}\triangleq e^{-\jmath\frac{4\pi f_c\Delta_\tau}{N_\mathrm{an}}}$, $W_\mathrm{T}\triangleq e^{\jmath\frac{2\pi N_\mathrm{d}N_\mathrm{S}}{N_\mathrm{do}N}}$. $\mathbf{F}_{N\times G}$ denotes the matrix composed of the first $G$ ($\leq N$) columns of $\mathbf{F}_N$. $N_{(\mathrm{S})}$ and $N_{(\mathrm{T})}$ are integers greater than or equal to $M+N_\mathrm{an}-1$ and $N_\mathrm{F}+N_\mathrm{do}-1$, respectively. $\boldsymbol{\xi}_{W,\alpha_1,\alpha_2,N}\in\mathbb{C}^N$ with $\left[\boldsymbol{\xi}_{W,\alpha_1,\alpha_2,N}\right]_i=W^{\left(i^2-\alpha_1 i-\alpha_2\right)/2}$, and $\tilde{\boldsymbol{\xi}}_{W,N_1,N_2,N}\triangleq\sqrt{N}\mathbf{F}_N\left[\boldsymbol{\xi}_{W,0,0,N_1}^\mathrm{H},\mathbf{0},\boldsymbol{\xi}_{W,2(N_2-1),-(N_2-1)^2,N_2-1}^\mathrm{H}\right]^\mathrm{T}\in\mathbb{C}^N$. Therefore, $[\mathbf{\Phi}_\mathrm{an}]_{:,0}$, $[\mathbf{\Phi}_\mathrm{de}]_{:,0}$ and $[\mathbf{\Phi}_\mathrm{do}]_{:,0}$ can be calculated efficiently. In conclusion, to construct the $\hat{\mathbf{P}}^\mathrm{H}\mathbf{T}^\mathrm{H}\hat{\mathbf{P}}$, we can just calculate and store $[\mathbf{\Phi}_\mathrm{an}]_{:,0}$, $[\mathbf{\Phi}_\mathrm{de}]_{:,0}$ and $[\mathbf{\Phi}_\mathrm{do}]_{:,0}$, whose computational complexity is $\mathcal{O}\left(N_{(\mathrm{S})}\log_2N_{(\mathrm{S})}+\bar{N}_\mathrm{de}\log_2\bar{N}_\mathrm{de}+N_{(\mathrm{T})}\log_2N_{(\mathrm{T})}\right)$ and storage complexity is $\mathcal{O}\left(N_\mathrm{an}+\bar{N}_\mathrm{de}+N_\mathrm{do}\right)$, which are much lower than that of calculating $\hat{\mathbf{P}}^\mathrm{H}\mathbf{T}^\mathrm{H}\hat{\mathbf{P}}$ directly.

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By exploiting the asymptotic orthogonality among TBs, we can further reduce the complexity of (46). We rewrite (46) as

$$\begin{aligned}&\hat{\mathbf{w}}_{\bar{n}}^{\text{turbo},(t)}\\&=\Big(\sigma_{\text{p}}^2\hat{\mathbf{Q}}_{\bar{n}}\Big(\big[\bar{\mathbf{R}}^{\text{TB}}\big]_{\bar{n},\bar{n}}\mathbf{e}_{\bar{n}}\mathbf{e}_{\bar{n}}^{\text{H}}+\sum_{\bar{n}'=0,\bar{n}'\neq\bar{n}}^{N_{\text{TB}}-1}\Big[\bar{\mathbf{R}}_{\text{turbo}}^{\text{TB},(t-1)}\Big]_{\bar{n}',\bar{n}'}\mathbf{e}_{\bar{n}'}\mathbf{e}_{\bar{n}'}^{\text{H}}\Big)\hat{\mathbf{Q}}_{\bar{n}}^{\text{H}}\\&\quad+\sigma_{\text{z}}^2\hat{\mathbf{B}}_{\bar{n}}^{\text{H}}\hat{\mathbf{P}}^{\text{H}}\mathbf{T}^{\text{H}}\mathbf{T}\hat{\mathbf{P}}\hat{\mathbf{B}}_{\bar{n}}\Big)^{-1}\hat{\mathbf{Q}}_{\bar{n}}\mathbf{e}_{\bar{n}}\big[\bar{\mathbf{R}}^{\text{TB}}\big]_{\bar{n},\bar{n}}\sigma_{\text{p}}.\end{aligned}\tag{55}$$

Due to the asymptotic orthogonality among the TBs (columns in $\bar{\mathbf{P}}$) as shown in Fig. 5 and Fig. 7 (when window functions are considered), we have

$$\begin{aligned}&\sum_{\bar{n}'=0,\bar{n}'\neq\bar{n}}^{N_{\text{TB}}-1}\Big[\bar{\mathbf{R}}_{\text{turbo}}^{\text{TB},(t-1)}\Big]_{\bar{n}',\bar{n}'}\hat{\mathbf{Q}}_{\bar{n}}\mathbf{e}_{\bar{n}'}\mathbf{e}_{\bar{n}'}^{\text{H}}\hat{\mathbf{Q}}_{\bar{n}}^{\text{H}}\\&\approx\sum_{\bar{n}'\in\mathcal{T}_{\bar{n}}}\Big[\bar{\mathbf{R}}_{\text{turbo}}^{\text{TB},(t-1)}\Big]_{\bar{n}',\bar{n}'}\hat{\mathbf{Q}}_{\bar{n}}\mathbf{e}_{\bar{n}'}\mathbf{e}_{\bar{n}'}^{\text{H}}\hat{\mathbf{Q}}_{\bar{n}}^{\text{H}},\end{aligned}\tag{56}$$

where $\mathcal{T}_{\bar{n}}=\big\{\bar{n}'\big|\big\|\hat{\mathbf{Q}}_{\bar{n}}\mathbf{e}_{\bar{n}'}\big\|_{\infty}>\varepsilon\big\}$, $\varepsilon$ is a threshold to filter out the TBs whose interference can be omitted. Thus (55) can be approximated as

$$\begin{aligned}&\hat{\mathbf{w}}_{\bar{n}}^{\text{turbo},(t)}\\&\approx\Big(\sigma_{\text{p}}^2\hat{\mathbf{Q}}_{\bar{n}}\Big(\big[\bar{\mathbf{R}}^{\text{TB}}\big]_{\bar{n},\bar{n}}\mathbf{e}_{\bar{n}}\mathbf{e}_{\bar{n}}^{\text{H}}+\sum_{\bar{n}'\in\mathcal{T}_{\bar{n}}}\Big[\bar{\mathbf{R}}_{\text{turbo}}^{\text{TB},(t-1)}\Big]_{\bar{n}',\bar{n}'}\mathbf{e}_{\bar{n}'}\mathbf{e}_{\bar{n}'}^{\text{H}}\Big)\hat{\mathbf{Q}}_{\bar{n}}^{\text{H}}\\&\quad+\sigma_{\text{z}}^2\hat{\mathbf{B}}_{\bar{n}}^{\text{H}}\hat{\mathbf{P}}^{\text{H}}\mathbf{T}^{\text{H}}\mathbf{T}\hat{\mathbf{P}}\hat{\mathbf{B}}_{\bar{n}}\Big)^{-1}\hat{\mathbf{Q}}_{\bar{n}}\mathbf{e}_{\bar{n}}\big[\bar{\mathbf{R}}^{\text{TB}}\big]_{\bar{n},\bar{n}}\sigma_{\text{p}},\end{aligned}\tag{57}$$

whose complexity is $\mathcal{O}\Big(\big(\hat{N}_{\text{TB},\bar{n}}\big)^2|\mathcal{T}_{\bar{n}}|+\big(\hat{N}_{\text{TB},\bar{n}}\big)^3\Big)$.

### B. Low-Complexity Implementation

After the design of the TB domain estimator, we apply it to the received signal to obtain channel estimates, which is referred to as the implementation step, i.e., step 5 in Algorithm 1. Thus the implementation complexity comes from (45) and mainly from the calculation of $\hat{\mathbf{y}}_{\bar{n}}^{(t)}$ in (47), which consists of two parts: the calculation of $\tilde{\mathbf{y}}$ and $\hat{\mathbf{Q}}_{\bar{n}}\Big(\bar{\mathbf{h}}_{\text{turbo}}^{\text{TB},(t-1)}-\Big[\bar{\mathbf{h}}_{\text{turbo}}^{\text{TB},(t-1)}\Big]_{\bar{n}}\mathbf{e}_{\bar{n}}\Big)$.

According to (14), $\tilde{\mathbf{y}}$ can be rewritten as

$$\tilde{\mathbf{y}}=\Big([\mathbf{D}]_{\mathcal{P},:}^{\text{H}}\otimes\bar{\mathbf{U}}^{\text{H}}\otimes\mathbf{V}^{\text{H}}\Big)\mathbf{T}^{\text{H}}\bar{\mathbf{y}}.\tag{58}$$

Since $\bar{\mathbf{U}}$ is a partial DFT matrix, and $\mathbf{V}$ and $[\mathbf{D}]_{\mathcal{P},:}$ are CZT matrices as shown in (54), the calculation of $\tilde{\mathbf{y}}$ can be implemented efficiently with the complexity $\mathcal{O}\big(N_{\text{v}}N_{\text{F}}N_{(\text{S})}\log_2N_{(\text{S})}+N_{\text{an}}N_{\text{F}}\bar{N}_{\text{de}}\log_2\bar{N}_{\text{de}}+N_{\text{an}}\bar{N}_{\text{de}}N_{(\text{T})}\log_2N_{(\text{T})}\big)$. Note that $\tilde{\mathbf{y}}$ needs to be calculated only once for the whole turbo-BSCE algorithm. Moreover, similar to (56), we have

$$\hat{\mathbf{Q}}_{\bar{n}}\Big(\bar{\mathbf{h}}_{\text{turbo}}^{\text{TB},(t-1)}-\Big[\bar{\mathbf{h}}_{\text{turbo}}^{\text{TB},(t-1)}\Big]_{\bar{n}}\mathbf{e}_{\bar{n}}\Big)\approx\hat{\mathbf{Q}}_{\bar{n}}\sum_{\bar{n}'\in\mathcal{T}_{\bar{n}},\bar{n}'\neq\bar{n}}\Big[\bar{\mathbf{h}}_{\text{turbo}}^{\text{TB},(t-1)}\Big]_{\bar{n}'}\mathbf{e}_{\bar{n}'},\tag{59}$$

whose complexity is $\mathcal{O}\big(\hat{N}_{\text{TB},\bar{n}}|\mathcal{T}_{\bar{n}}|\big)$.

In conclusion, the complexity of the whole turbo-BSCE, i.e., the sum of the design complexity and the implementation complexity, is $\mathcal{O}\Big(\Big(\sum_{\bar{n}=0}^{\bar{N}_{\text{TB}}-1}\big(\hat{N}_{\text{TB},\bar{n}}\big)^2|\mathcal{T}_{\bar{n}}|+\big(\hat{N}_{\text{TB},\bar{n}}\big)^3\Big)T+N_{\text{v}}N_{\text{F}}N_{(\text{S})}\log_2N_{(\text{S})}+N_{\text{an}}N_{\text{F}}\bar{N}_{\text{de}}\log_2\bar{N}_{\text{de}}+N_{\text{an}}\bar{N}_{\text{de}}N_{(\text{T})}\log_2N_{(\text{T})}\Big)$.

TABLE I
SIMULATION PARAMETERS

| Parameter | Value |
|---|---|
| Carrier frequency $f_{\text{c}}$ | 16 MHz |
| Subcarrier spacing $\Delta_f$ | 125 Hz |
| Number of subcarriers $N_{\text{c}}$ | 2048 |
| Length of CP $N_{\text{g}}$ | 512 |
| Number of valid subcarriers $N_{\text{v}}$ | 1536 |
| Number of BS antennas $M$ | 128 |
| BS antenna spacing $d$ | 9 m |
| Size of frame $(N_{\text{F}},N_{\text{S}},n_{\text{p}})$ | (8, 14, 6) |
| Number of UTs $U$ | 64 |
| Velocity of UTs $v$ | 100 km/h |

Due to the channel sparsity in the TB domain, the number of designed TB domain estimators, $\bar{N}_{\text{TB}}$, is typically not large and the above complexity is much lower than that of the MMSE estimation.

## V. SIMULATION RESULTS

In this section, we provide simulation results to illustrate the efficiency of the proposed channel estimation approach for HF skywave massive MIMO-OFDM communications.

### A. Simulation Setup

The main simulation parameters are given in Table I. UTs are generated at a distance of 2,000 km from the BS and the azimuth angles of UTs seen from the BS are generated in the interval (-90°,90°). To obtain realistic HF skywave channels, the channel parameters of each UT are generated independently using a commercial ray-tracing software Problab-Pro version 3.1 [52]. Specifically, the software is utilized to obtain the parameters of each propagation path between the BS and each UT, including the path gain, the propagation delay, the azimuth and elevation AoA. Note that in HF skywave communications, the Doppler frequency comes from the ionosphere and UT mobility. Thus the maximum Doppler is $\nu_{\max}=\frac{\nu_{\text{iono}}}{2}+\frac{v}{c}f_{\text{c}}$, where $\nu_{\text{iono}}$ is the Doppler spread caused by the ionosphere and $v$ is the velocity of UTs. We set $\nu_{\text{iono}}$ to 0.5 Hz for moderate ionospheric conditions at mid-latitude regions [53] and randomly generate the Doppler frequency of each path within the range $[-\nu_{\max},\nu_{\max}]$. $N_{\text{d}}$ is set to be 8 according to the maximum Doppler. With the generated channel parameters, we can obtain the SFT domain channel of each UT according to (1), where fine factors, $F_{\text{an}}$, $F_{\text{de}}$ and $F_{\text{do}}$, are set to two [35], and the corresponding statistical CSI can be acquired by the method in [54].

The pilot allocation for each UT is performed by a similar approach in [35] to make the equivalent TB domain channels of all UTs non-overlapping. Moreover, we define the total transmit power of all UTs as $P_{\text{t}}=U\sigma_{\text{p}}^2$ and the noise power spectral density at the BS is set to -163 dBW/Hz [55].

### B. Channel Estimation Performance

Fig. 9 shows the NMSE performance of BSCE with different numbers of selected TBs, where the window functions are the rectangular windows, $\varepsilon=0$ and no iterations are used (i.e., $T=1$ for turbo-BSCE). We select TBs used for BSCE in the

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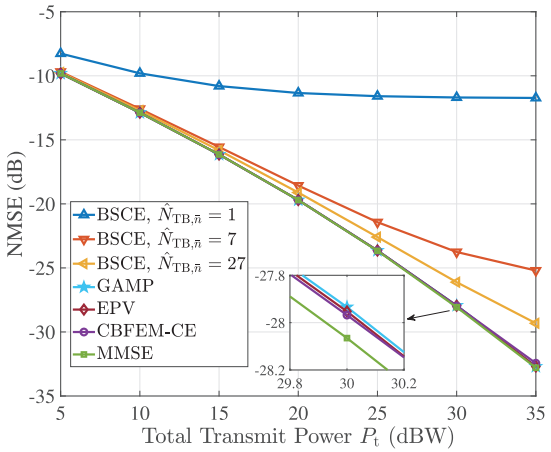



Fig. 9. NMSE performance of BSCE with different numbers of selected TBs compared with other channel estimation algorithms.

way shown in Fig. 6. $\hat{N}_{\mathrm{TB},\bar{n}} = 1$ means that we only select the TB that each TB domain channel element occupies, $\hat{N}_{\mathrm{TB},\bar{n}} = 7$ means that we select TBs in a cross pattern as shown on the left of Fig. 6, and $\hat{N}_{\mathrm{TB},\bar{n}} = 27$ means that we select TBs in a cubic pattern as shown on the right of Fig. 6. Besides the MMSE estimation, the constrained Bethe free energy minimization based channel estimation algorithm (CBFEM-CE) proposed in [35], the generalized approximate message passing (GAMP) [56] and the expectation propagation variant (EPV) [57] algorithms are included for comparison. Note that all these reference algorithms leverage the statistical CSI to estimate the channel. From the figure, the performance of the proposed BSCE can approach that of the MMSE estimation and other compared algorithms. Even though the performance gap increases with the transmit power, the gap can be reduced by including a higher number of TBs. This is because the performance of BSCE can be further improved by using more TBs in addition to the one occupied by each TB domain channel element itself to tackle the inter-TB interference.

Fig. 10 shows the NMSE performance of BSCE with different window functions and different thresholds $\varepsilon$, where $T = 1$ and $\hat{N}_{\mathrm{TB},\bar{n}} = 7$. From the figure, the proposed BSCE with different window functions have a similar performance. Moreover, filtering out the TBs whose interference can be omitted according to an appropriate threshold value has a negligible impact on performance. Note that compared with the rectangular window, the cosine window reduces the side-lobe amplitude but has a wider main-lobe. It means that when we design the channel estimator for a certain TB domain channel element, although the cosine window can mitigate the interference from the other TB domain channel elements located outside the main-lobe, it may enlarge the interference from those located inside the main-lobe, thus the final performance can not be guaranteed to be better. However, from Fig. 11, the number of reserved TBs of the cosine window after filtering out the TBs with negligible interference is less than that of the rectangular window under the same threshold, $\varepsilon$, especially when $\varepsilon$ is small. This is since the cosine window can make the TB domain energy more concentrated compared with the rectangular window as shown in Fig. 5 and Fig. 7, which leads to a lower complexity.

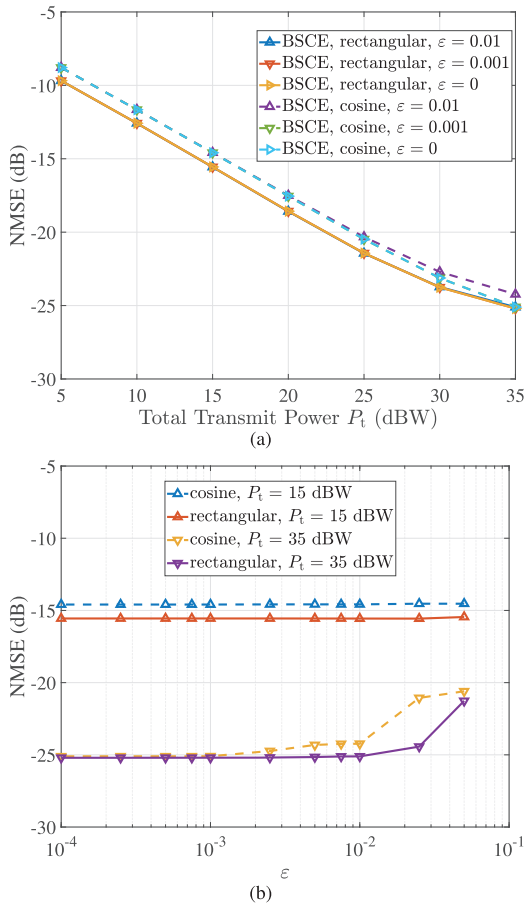



Fig. 10. NMSE performance of BSCE with different window functions versus (a) total transmit power $P_t$; (b) threshold $\varepsilon$.

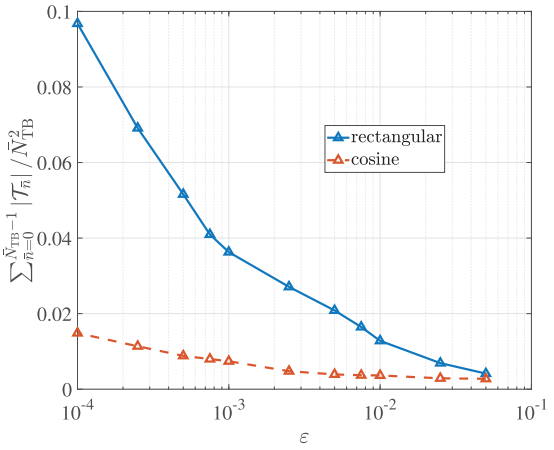



Fig. 11. The ratio between the average number of reserved TBs after filtering out the TBs whose interference is negligible, i.e., $\sum_{\bar{n}=0}^{\bar{N}_{\mathrm{TB}}-1} |\mathcal{T}_{\bar{n}}| / \bar{N}_{\mathrm{TB}}$, and the total number of non-zero TBs $\bar{N}_{\mathrm{TB}}$, versus threshold $\varepsilon$, where $\hat{N}_{\mathrm{TB},\bar{n}} = 7$.

Fig. 12 shows the NMSE performance of turbo-BSCE, where the window functions are the rectangular windows, $\hat{N}_{\mathrm{TB},\bar{n}} = 7$ and $\varepsilon = 0.001$. From the figure, the NMSE performance of turbo-BSCE is improved significantly over the iterations. Moreover, the performance of turbo-BSCE can approach that of the MMSE estimation in just a few iterations even if the number of selected TBs, $\hat{N}_{\mathrm{TB},\bar{n}}$, is not large, while the reference algorithms require much more iterations.

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TABLE II
COMPUTATIONAL COMPLEXITY OF DIFFERENT CHANNEL ESTIMATION ALGORITHMS

| Algorithm | Complexity |
|---|---|
| MMSE | $\mathcal{O}\left(\bar{N}_{\mathrm{TB}}^2 M N_{\mathrm{v}} N_{\mathrm{F}} + \bar{N}_{\mathrm{TB}}^3\right)$ |
| GAMP / EPV | $\mathcal{O}\left(M N_{\mathrm{v}} N_{\mathrm{F}} \bar{N}_{\mathrm{TB}} T\right)$ |
| CBFEM-CE | $\mathcal{O}\Big(\left(\left(N_{\mathrm{an}} N_{\mathrm{do}} + N_{\mathrm{an}} N_{\mathrm{F}}\right) \bar{N}_{\mathrm{de}} \log_2 \bar{N}_{\mathrm{de}} + \left(N_{\mathrm{v}} N_{\mathrm{do}} + N_{\mathrm{v}} N_{\mathrm{F}}\right) N_{(\mathrm{S})} \log_2 N_{(\mathrm{S})} + \left(M N_{\mathrm{v}} + N_{\mathrm{an}} \bar{N}_{\mathrm{de}}\right) N_{(\mathrm{T})} \log_2 N_{(\mathrm{T})}\right) T\Big)$ |
| Turbo-BSCE | $\mathcal{O}\Big(\Big(\sum_{\bar{n}=0}^{\bar{N}_{\mathrm{TB}}-1} \left(\hat{N}_{\mathrm{TB},\bar{n}}\right)^2 \lvert\mathcal{T}_{\bar{n}}\rvert + \left(\hat{N}_{\mathrm{TB},\bar{n}}\right)^3\Big) T + N_{\mathrm{v}} N_{\mathrm{F}} N_{(\mathrm{S})} \log_2 N_{(\mathrm{S})} + N_{\mathrm{an}} N_{\mathrm{F}} \bar{N}_{\mathrm{de}} \log_2 \bar{N}_{\mathrm{de}} + N_{\mathrm{an}} \bar{N}_{\mathrm{de}} N_{(\mathrm{T})} \log_2 N_{(\mathrm{T})}\Big)$ |

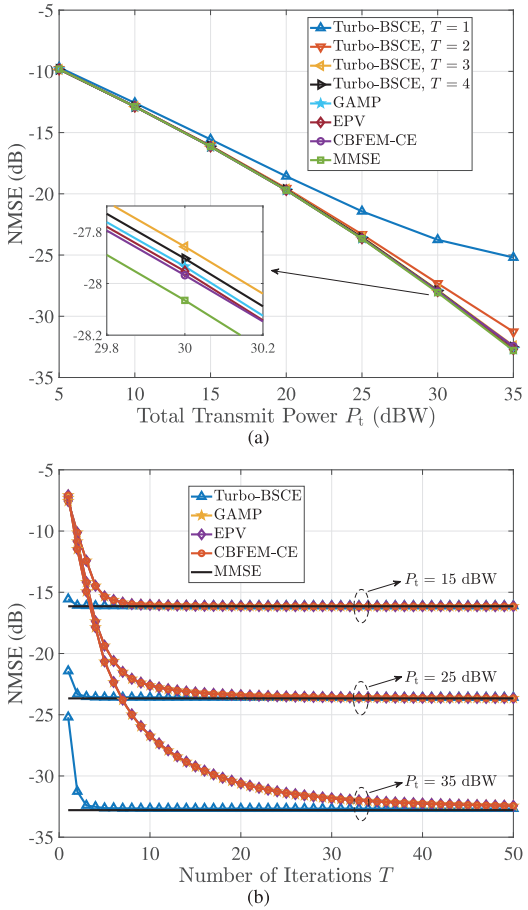



Fig. 12. NMSE performance of turbo-BSCE and other channel estimation algorithms versus (a) total transmit power $P_{\mathrm{t}}$; (b) number of iterations $T$.

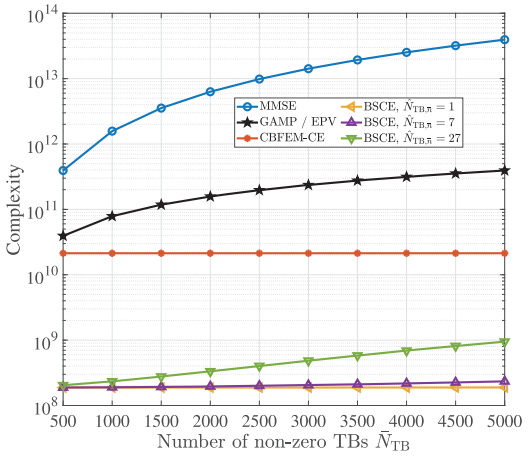



Fig. 13. Complexity of BSCE with different numbers of selected TBs compared with other channel estimation algorithms.

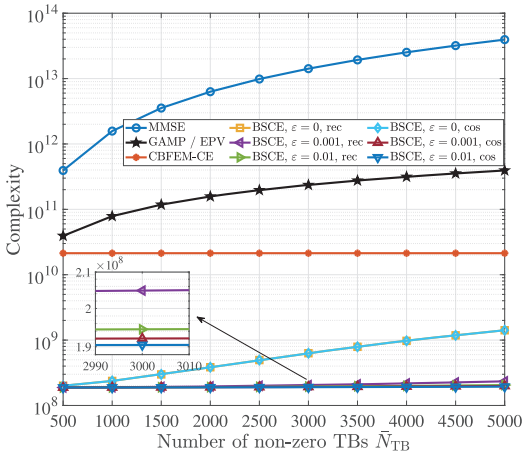



Fig. 14. Complexity of BSCE with different window functions and different thresholds compared with other channel estimation algorithms.

Therefore, the proposed turbo-BSCE can achieve a good trade-off between performance and complexity.

### C. Complexity

The computational complexity in terms of the number of complex multiplications of different channel estimation algorithms are summarized in Table II and compared in Fig. 13–15.

Fig. 13 shows the complexity of BSCE with different numbers of selected TBs, when using rectangular window functions, $T = 1$ and $\varepsilon = 0.001$, compared with other algorithms. The numbers of iterations for CBFEM-CE, GAMP and EPV algorithms are set to 50. From the figure, the complexity of BSCE increases as the number of selected TBs, $\hat{N}_{\mathrm{TB},\bar{n}}$, increases. However, by exploiting the TB domain channel sparsity and the beam structure, the complexity of BSCE is much lower than that of the MMSE estimation and other compared algorithms.

Fig. 14 shows the complexity of BSCE with different window functions and different thresholds, where $T = 1$ and $\hat{N}_{\mathrm{TB},\bar{n}} = 7$, compared with other algorithms. From the figure, a proper threshold $\varepsilon$ can help remove the TBs with negligible interference, thus significantly reduce the complexity while ensuring a negligible performance loss, as shown in Fig. 10. Moreover, BSCE with the cosine window achieves a lower complexity compared with BSCE with the rectangular window under the same threshold $\varepsilon$ since the number of selected TBs with the cosine window is less than that with the rectangular window as shown in Fig. 11.

Fig. 15 shows the complexity of turbo-BSCE with different numbers of iterations, when using rectangular window functions, $\hat{N}_{\mathrm{TB},\bar{n}} = 7$ and $\varepsilon = 0.001$, compared with other

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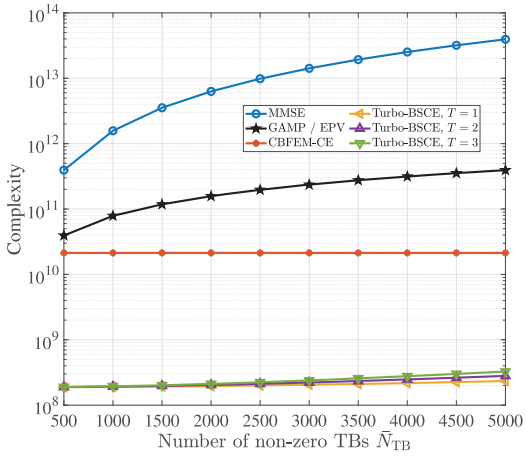



Fig. 15. Complexity of turbo-BSCE with different numbers of iterations compared with other channel estimation algorithms.

algorithms. From the figure, turbo-BSCE has a lower complexity than the MMSE estimation and other compared algorithms due to the utilization of the channel sparsity in the TB domain and the much lower number of iterations. Moreover, a few iterations will not greatly increase the complexity of turbo-BSCE as shown in Fig. 15 but will significantly improve its performance as shown in Fig. 12. Hence, the proposed turbo-BSCE can reduce the complexity while ensuring a similar performance to the MMSE estimation.

## VI. Conclusion

In this paper, we have investigated the channel estimation for HF skywave massive MIMO-OFDM communications. We first reviewed the TB based channel model and the channel sparsity in the TB domain. With the TB based channel model, we formulated the channel estimation for all UTs as an estimation of the sum of equivalent TB domain channels for all UTs, from which the TB domain channels of all UTs can be acquired. Then we proposed BSCE, where we showed that the SFT domain estimator can be designed by optimizing a low-dimensional TB domain estimator and the way to choose TBs for BSCE was also given. Moreover, we introduced window functions and the turbo principle into BSCE to generalize it, thus making a good trade-off between complexity and performance. Furthermore, we presented a low-complexity design and implementation of BSCE by exploiting the characteristics of the TB matrix. Simulation results verified the effectiveness of our proposed channel estimation approach.

## Appendix
## Proof of Lemma 1

Define $\tilde{\mathbf{w}}_{\bar{n}} = \bar{\mathbf{P}}^{\mathrm{H}}\mathbf{w}_{\bar{n}} \in \mathbb{C}^{\bar{N}_{\mathrm{TB}}}$, (19) is rewritten as

$$\mathrm{MSE}_{\bar{n}} = \left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}} - \sigma_{\mathrm{p}}\left[\bar{\mathbf{R}}^{\mathrm{TB}}\right]_{\bar{n},\bar{n}}\left(\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{H}}\mathbf{e}_{\bar{n}} + \mathbf{e}_{\bar{n}}^{\mathrm{H}}\tilde{\mathbf{w}}_{\bar{n}}\right) + \sigma_{\mathrm{p}}^2\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{H}}\bar{\mathbf{R}}^{\mathrm{TB}}\tilde{\mathbf{w}}_{\bar{n}} + \sigma_{\mathrm{z}}^2\mathbf{w}_{\bar{n}}^{\mathrm{H}}\mathbf{w}_{\bar{n}}. \quad (60)$$

In (60), $\mathbf{w}_{\bar{n}}$ in most terms is replaced by $\tilde{\mathbf{w}}_{\bar{n}}$. Although $\mathbf{w}_{\bar{n}}$ is still involved in the term $\sigma_{\mathrm{z}}^2\mathbf{w}_{\bar{n}}^{\mathrm{H}}\mathbf{w}_{\bar{n}}$, it can be obtained according to the relation $\tilde{\mathbf{w}}_{\bar{n}} = \bar{\mathbf{P}}^{\mathrm{H}}\mathbf{w}_{\bar{n}}$ when $\tilde{\mathbf{w}}_{\bar{n}}$ is given. Therefore, to ensure a proper of $\mathbf{w}_{\bar{n}}$ such that $\tilde{\mathbf{w}}_{\bar{n}} = \bar{\mathbf{P}}^{\mathrm{H}}\mathbf{w}_{\bar{n}}$ when $\tilde{\mathbf{w}}_{\bar{n}}$ is given, $\tilde{\mathbf{w}}_{\bar{n}}$ should satisfy the condition $\left(\bar{\mathbf{P}}^{\mathrm{H}}\left(\bar{\mathbf{P}}^{\mathrm{H}}\right)^{\dagger} - \mathbf{I}\right)\tilde{\mathbf{w}}_{\bar{n}} = \mathbf{0}$ [58]. Therefore, we can turn to optimize $\tilde{\mathbf{w}}_{\bar{n}}$ instead of $\mathbf{w}_{\bar{n}}$ to minimize the MSE. Moreover, note that for a given optimal solution $\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}$, the corresponding optimal estimator $\mathbf{w}_{\bar{n}}^{\mathrm{op}}$ should have the minimal norm, i.e., $\left(\mathbf{w}_{\bar{n}}^{\mathrm{op}}\right)^{\mathrm{H}}\mathbf{w}_{\bar{n}}^{\mathrm{op}}$ should be minimal, and we can obtain the corresponding $\mathrm{MSE}_{\bar{n}}$ in (60). Otherwise, we will have a $\mathbf{w}_{\bar{n}}^{\mathrm{op}}$ with lower norm, which leads to a lower $\mathrm{MSE}_{\bar{n}}$ and contradicts the optimality. Therefore, we can formulate the following optimization problem

$$\mathbf{w}_{\bar{n}}^{\mathrm{op}} = \arg\min_{\mathbf{w}_{\bar{n}}} \ \mathbf{w}_{\bar{n}}^{\mathrm{H}}\mathbf{w}_{\bar{n}}, \quad \text{s.t. } \bar{\mathbf{P}}^{\mathrm{H}}\mathbf{w}_{\bar{n}} = \tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}. \quad (61)$$

The optimal solution of problem (61) can be obtained as

$$\mathbf{w}_{\bar{n}}^{\mathrm{op}} = \bar{\mathbf{P}}\left(\bar{\mathbf{P}}^{\mathrm{H}}\bar{\mathbf{P}}\right)^{\dagger}\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}. \quad (62)$$

Then according to (62), we replace $\mathbf{w}_{\bar{n}}$ with $\bar{\mathbf{P}}\left(\bar{\mathbf{P}}^{\mathrm{H}}\bar{\mathbf{P}}\right)^{\dagger}\tilde{\mathbf{w}}_{\bar{n}}^{\mathrm{op}}$ in (60) and obtain the objective function in (22). This completes the proof.

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